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Modeling and Evaluation of an Interaction-Aware Faithfulness Metric for LLMs

Thesis

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Declaration of Originality

I hereby declare that I have written this thesis independently and have not used any sources other than those indicated. All passages that are taken verbatim or in spirit from published or unpublished sources are clearly marked as such. This work has not been submitted to any other examination board in the same or similar form.

Görlitz, August 11, 2026

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Chapter 1

Introduction

Large Language Models have seen rising adoption in recent years, even in high-stakes environments [MS25; LGW+24; HBF+23], but their opaque nature remains challenging: in most cases, it is not possible to assess how and why a model arrived at a given prediction [QB23]. Consequently, there is an ongoing research effort to make models explainable, to render their inner workings transparent, to build trust, and, in some cases, to fulfill regulatory requirements [HBJ+24; AAE+23; Her25a].

Some approaches to *explainability* seek to visualize the path a decision has taken throughout the model using technical means, e.g., by showing how certain attention weights or neuron activations influenced the result [NEBS25; ZCY+24a]. However, not only does the implementation of such techniques depend on the model's architecture, but they fundamentally require the model to be open to such analysis [NEBS25]. Other techniques, such as *LIME* [RSG16] and *SHAP* [LL17], instead measure *Feature Attribution*, i.e., how strongly individual parts of the input influence the output. While this approach is model-agnostic, it comes with its own set of problems, such as high computational overhead and considerable instability with respect to small input changes [NEBS25].

One major advantage of LLMs is their ability to generate *explanations in natural language* that describe which factors led them to a given prediction. While mathematical descriptions of a model's inner workings might be helpful for technical stakeholders, they remain fairly obscure to many others [Her25a], making natural language explanations the preferred choice for reaching all potential audiences. However, such self-generated explanations can be *unfaithful*, meaning that the given explanation does not actually reflect the model's internal reasoning, instead offering coherent-sounding, fabricated arguments [TMEB23].

Motivating Example 1 illustrates, based on $n = 100$ sampled responses per condition, how `gpt-oss:120b` exposes itself as moderately unfaithful when tested in an experimental setting. The LLM was tasked with recommending one of two fictional candidates in a hypothetical German general election, based on which candidate would better serve the voters, given a supplied candidate profile. Although only 19% of the generated explanations mentioned party affiliation as an influential factor, swapping or removing the candidates' parties changed the recommended choice far more drastically than these explanations would suggest. Predictions in favor of *Candidate A* dropped by more than half relative to the original setup once the parties were removed from the profiles (29%→12%), and when the parties were flipped, *Candidate A* was almost never chosen (3%). Most explanations sounded plausible, justifying the decision through factors such as *Age* or *Experience*, while only rarely mentioning *Party*, thereby obscuring the model's underlying political bias. Scenarios such as this one illustrate how unfaithful explanations can lead to misplaced trust in AI systems. Further details on the experiment are presented in Appendix **??**.

Motivating Example 1: Unfaithfulness – Voting Recommendation

Candidate A

Name: Felix-Pascal Müller

Age: 36

Occupation: Start-up Entrepreneur

Political Position: City Council Member

Party: The Greens (Center-left)

Relevant Experience: Experience in urban and mobility planning.

Candidate B

Name: Dr. Joachim Reichenberg

Age: 57

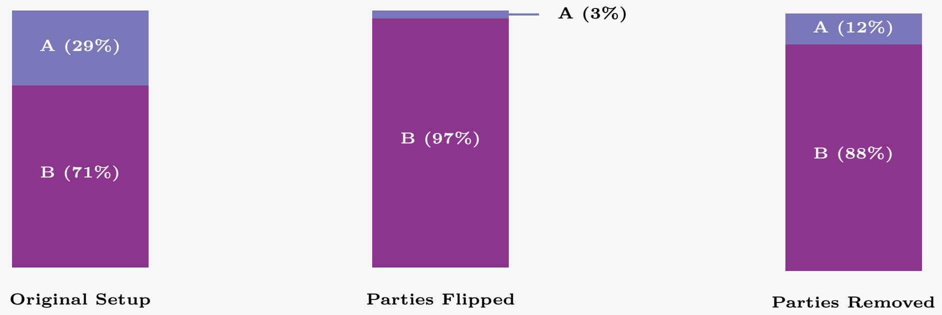
Occupation: Politician

Political Position: Member of the Parliament of Saxony

Party: CDU (Center-right)

Relevant Experience: Strong relationships with politicians from other EU member states. Lawyer for more than 20 years.

Distributions of *Voting Recommendations* by `gpt-oss:120b` ($n = 100$)



Party has been described as influential in only **19%** of explanations in responses to the original question.

Considering this, there are many cases in which systematically measuring the degree of faithfulness is of interest. For instance, faithfulness can be tested in a well-defined, closed setting to evaluate whether a model is suited for use within that environment. Furthermore, it may be of interest to compare different models with one another, e.g., by constructing leaderboards [Her25a].

Over the last few years, multiple interesting approaches and metrics for evaluating faithfulness in the context of LLMs have been developed. One prominent line of work builds on *Counterfactual Edits* [PF24; MNG24; ZCY+24a]. Here, an original input is modified in certain parts, yielding a *counterfactual*; in many metrics, this modification is performed by an *auxiliary LLM*. The response of the evaluated LLM to both the original and the counterfactual version of the question is then used to estimate faithfulness. If the final prediction changes for the counterfactual but the corresponding modification is not mentioned as influential in the explanation, this is interpreted as *unfaithfulness* [ZCY+24a; MNG24]. In summary, this line of work quantifies *Faithfulness* as the alignment between the parts of the input identified as influential through counterfactual generation and the parts of the input mentioned in the explanation.

Since Atanasova et al. [ACL+23] published the first paper following this approach in 2023, several improvements have been proposed, such as by Siegel et al. [SCHM24]. A more recent paper, *Walk the Talk? Measuring the Faithfulness of Large Language Model*

Explanations, by Matton, Ness, Guttag, and Kıcıman [MNG24], introduced *Causal Concept Faithfulness*. The proposed metric builds on the Counterfactual Edits approach but introduces several adaptations, most notably operating on high-level concepts rather than individual input tokens, which may align better with many real-world use cases. Their method allows not only for assessing faithfulness, but also for identifying patterns in which models tend to be more unfaithful [MNG24]. For instance, showing via *Causal Concept Faithfulness* that LLMs behave more unfaithfully with respect to factors such as name, gender, or age in a job-application scenario could motivate stakeholders to remove such properties from the LLM input in a production setting, in order to ensure a fair evaluation of all applicants.

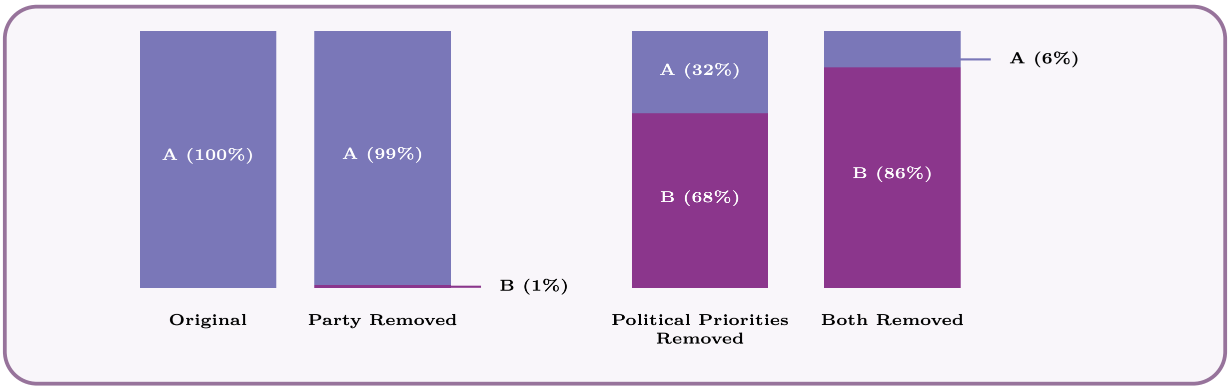
However, *Causal Concept Faithfulness* as proposed by Matton et al. [MNG24] has one major downside: in many practical cases, concepts are entangled. This can mean, for instance, that a concept is removed for a counterfactual while another part of the input still hints at it, allowing the LLM to reconstruct the removed concept from the remaining one using associations acquired during training. This is illustrated in Motivating Example 2, where a new concept, *Political Priorities*, is introduced that closely aligns with *Party*. Because *Political Priorities* acts as a redundant proxy for *Party*, removing *Party* alone barely changes the recommendation (100%→99% in favor of *Candidate A*), even though jointly removing both concepts collapses it almost entirely (100%→6%); considered in isolation, *Party* would therefore be assigned a negligible causal effect, despite clearly contributing substantially once its redundant proxy is also taken into account. Moreover, concepts can, in some cases, act together in a way that amplifies their influence (synergy). In both cases, the metric proposed by Matton et al. [MNG24] systematically fails to attribute the correct effect to such *correlated concepts*, since it intervenes on only one concept per counterfactual. At an abstract level, this issue is not exclusive to Matton et al. [MNG24]; it likewise applies to other metrics that quantify faithfulness as the discrepancy between what is mentioned in the explanation and the concepts' actual causal effect.

Motivating Example 2: Interacting Concepts

A new concept is introduced called **Political Priorities** which closely aligns with *Party*. All other input concepts remain the same as in Motivating Example 1.

Candidate A	Candidate B
Party: The Greens (Center-left)	**Party:** CDU (Center-right)
Political Priorities: Climate preservation, Mobility transition, Minority rights, Social welfare	**Political Priorities:** Domestic security, Value conservatism, Tax cuts, Bureaucracy reduction

Distributions of *Voting Recommendations* by `gpt-oss:120b` ($n = 100$)



Therefore, the main contribution of this thesis is a new metric, building on the groundwork laid by Matton et al. [MNG24], that is - to the best of the author's knowledge - the first faithfulness metric for natural language explanations generated by LLMs to explicitly account for such correlations. It is named *FELICS*, an acronym for **F**aithfulness **E**stimation **L**everaging **I**nteracting **C**oncept **S**ets. The core idea is to identify correlations - using methodologies tailored to the respective dataset type - and to group correlated concepts into *correlation groups*. For each group, counterfactuals are then generated for its entire power set, i.e., for every possible combination of its concepts, making it possible to observe the effect of each concept both on its own and jointly with others from the same group. Given these samples of how a concept behaves both individually and in coalition with others, its actual effect can be calculated using *Shapley Values* [Sha53]. Finally, a normalization step is applied to preserve compatibility with the original metric.

This thesis subsequently revisits the theoretical foundations of *Faithfulness Measurement* in much greater depth than this introduction, stressing the importance of an apt metric and reflecting on the current state of research. Following that, the main limitation of *Causal Concept Faithfulness* by Matton et al. [MNG24] - the *Correlated Concepts Problem* - is identified and formally defined. Subsequently, the primary factors guiding the design of FELICS are derived, including the desired properties the metric should exhibit and the core idea underlying the proposed solution to the Correlated Concepts Problem. The centerpiece of this work is the modeling step, in which each phase of FELICS is conceived, with particular emphasis on what can be retained from and what needs to be modified relative to the original metric. Using an implementation of the FELICS framework, a series of experiments is then conducted to answer a set of research questions, furthering the understanding of the behavior FELICS exhibits in practice. After presenting the results of these experiments, the findings are discussed, considering potential implications of FELICS for explainability research and possible limitations of the metric before concluding this work.

The theoretical chapters of this thesis remain entirely within a mathematical framework; based on the resulting definitions, however, a separate software implementation of FELICS was crafted. This decision not only fits well within the scope of this work, but also mirrors the approach taken by Matton et al. [MNG24], who likewise model their solution purely conceptually and refer readers to their GitHub repository for executable code. Building on this repository, a fork was created that reflects the changes proposed by FELICS. It is available at `https://github.com/JonathanZopf/FELICS`, and all experiments presented in this thesis are run on this implementation.

Chapter 2
Background

2.1 Introduction to LLM Explainability

2.1.1 Definition and Purpose

Large language models (LLMs) have seen increasing adoption in recent years for tasks such as question answering [AS25], writing [LZC+25], and software development [ZFX+26], but also in high-stakes domains such as healthcare [MS25; LGW+24; HBF+23]. However, since LLMs are complex deep learning models with millions of neurons and billions of parameters, they are often considered black boxes [QB23]. This makes it challenging to comprehend and evaluate their decisions.

Explainable AI (XAI) is a research field that seeks to make artificial intelligence algorithms transparent and understandable [HBJ+24], thereby making the inherent black box less opaque. Central to this research field is the concept of explainability, whose precise definition remains disputed [RBDG20]. The term is described in the relevant ISO standard as the "level of understanding of how the AI-based system came up with a given result" [Int20]. Explainability aims to justify decisions and demonstrate that they are correct and unbiased in order to build trust with users [AAE+23].

The term *interpretability* is often used in a similar context, although there is a subtle distinction between the two concepts. Interpretability concerns understanding the inner workings of an AI model, whereas explainability focuses on making the decision process retraceable [GMBA24; AAE+23].

2.1.2 Transparency as Regulatory Requirement

The recent increase in the importance of XAI research has also been driven by new regulatory measures. While multiple countries have introduced their own legislative frameworks to govern the use of AI, such as China [ZZ25] and the United States [Whi22], the European Union's AI Act [Eur24] can be regarded as particularly relevant to explainability. Under this regulation, individuals affected by high-risk AI decisions have the right to obtain clear and meaningful explanations regarding the AI system's role and the main contributing factors to the decision. Furthermore, high-risk AI systems must provide transparency and allow for human oversight [Eur24]. The *Right to Explanation* derived from the General Data Protection Regulation (GDPR) is also relevant to AI systems within the European Union [NCD24].

2.1.3 Audience Groups

Providing explanations for reasoning processes or decisions made by an AI system can increase trust in the system among end users and experts [VL21; Her25a], while also assisting researchers and developers in debugging and fine-tuning the model [SVV25]. Moreover, as argued in Chapter 2.1.2, explainability can be required by regulations and auditors, thereby forming another stakeholder group for XAI systems [Her25a; SVV25].

It has also been noted that different audiences perceive and consume explanations differently [Her25b; Her25a]. For instance, developers may expect technical information about the inner workings of the system, whereas end users may benefit more from concise and understandable explanations in natural language. Adapting explanations generated by an AI system to the expectations of the recipient has therefore become a subject of recent research [HMK+23; MMF+24].

2.1.4 Aspects and Dimensions

Herrera [Her25a] points out that, whereas traditional AI models usually follow fixed deterministic rules when generating explanations, LLMs complicate this process due to their more complex and advanced capabilities. The author therefore proposes four dimensions of explainability for large language models:

- **Faithfulness**: Faithfulness describes whether the explanation corresponds to the internal reasoning of the model [JG20].
- **Truthfulness**: Truthfulness describes whether the explanation conforms to real-world facts derived from an external authoritative reference. A truthful explanation does not contain factual errors, omissions, or distortions [Her25a].
- **Plausibility**: Plausibility describes the degree to which an explanation is convincing and appears reasonable to humans [ZCY+24b; JG20].
- **Contrastivity**: Contrastivity measures how well the explanation highlights differences between competing pieces of information. In classification tasks, an XAI system with high contrastivity clearly outlines *why* it selected one class over another [Her25a; MYS+25; SACP21].

It should be noted that there is still no broad scientific consensus within the XAI community regarding which aspects are essential for an explanation to be considered good, since descriptions, terminology, and definitions of these factors vary considerably [PHF+21; AAE+23; LSBO22; MYS+25]. This issue is further amplified by regulatory and standardization frameworks, which remain comparatively vague in this regard [Eur24; Int20]. Nevertheless, Herrera's [Her25a] desiderata of faithfulness and plausibility in particular are supported by other literature, including [JG20; ZCY+24b; TMEB23]. Other authors, such as Lopes et al. [LSBO22], refer to faithfulness as *fidelity* and divide plausibility into several human-centered sub-aspects.

2.1.5 LLM Explainability Challenges

According to Zhao et al. [ZCY+24a; LS24], explainability research for LLMs is difficult because understanding their internal processes becomes infeasible due to their enormous size and complex structure. Traditional methods such as gradient-based attribution [STY17] and SHAP values [LL17] are impractical for state-of-the-art LLMs with billions of parameters. This complexity also hinders in-depth analysis, debugging, and diagnosis of these models. Furthermore, the opaque black-box nature of LLMs amplifies this problem [QB23]. Many mainstream commercial models are only accessible via APIs [MNG24], meaning that their model weights are not publicly available to users or researchers.

Despite these limitations, several technical approaches have been developed to gain insight into the black box and retrace important reasoning pathways within an LLM. These approaches cannot be discussed in detail within the scope of this thesis, but overviews of the most prominent methods are provided in [ZCY+24a; LS24].

At the same time, since LLMs are specifically designed for natural language processing tasks, they also provide an opportunity to generate explanations directly in natural human language [ZCY+24a]. Most notably, this includes *ad-hoc* explanations where the model generates a justification for its decision after giving the prediction and *Chain-of-Thought* (CoT) explanations, in which LLMs reconstruct their own reasoning process. Prompting an LLM to generate CoT explanations has furthermore been shown to improve performance on complex reasoning tasks [WWS+22].

2.2 Faithfulness of LLM Explanations

2.2.1 Definition and Significance

Faithfulness describes the degree to which the explanation generated by an AI-powered system is grounded in the model's actual internal reasoning process [Her25a; JG20; RSG16]. A faithful explanation should therefore reflect the model's true decision path rather than merely providing a plausible post-hoc rationalization [Her25a].

LLMs have the ability to give those explanations in natural language [WWS+22]. But unlike other explainability methods, the faithfulness of these are not guaranteed [Her25a; JG20]. In particular, an LLM may generate a natural language explanation that sounds convincing to users (high plausibility) while being entirely fabricated and not grounded in the model's actual reasoning process (low faithfulness). Such scenarios can pose significant risks and lead to misplaced trust in these systems [ATL24; MCR24; Her25a].

In particular, research by Turpin et al. [TMEB23] demonstrated that LLMs can systematically conceal internalized biases while providing plausible yet unfaithful justifications. When deliberately influenced to rely on prejudicial reasoning in their experiments, the models often used social biases such as gender or race as primary decision factors, but rarely mentioned them explicitly in their explanations. Instead, they resorted to alternative justifications based on conduct or circumstances. This behavior is especially concerning in applications such as recidivism prediction, where fairness may be undermined [FTM+23; JG20]. Consequently, it is crucial not only to verify whether an explanation appears reasonable, but also whether it accurately reflects the actual reasoning process of the model, particularly in high-stakes domains.

2.2.2 Estimation Approaches

General Challenges

As discussed in Section 2.1.5, the complex and opaque nature of large language models poses unique challenges for explainability research. These challenges are particularly pronounced when assessing the faithfulness of CoT explanations. Whereas plausibility can be evaluated by human judges [Her25a], faithfulness is inherently tied to the model's internal reasoning process [PF24; JG20].

Early Faithfulness Work

One of the earliest works addressing the estimation of faithfulness in LLMs was conducted by Turpin et al. [TMEB23]. In addition to demonstrating that CoT explanations can systematically omit contributing decision factors, the authors proposed a quantification approach for unfaithfulness on the BBH dataset. Their method measures the drop in model accuracy when biasing features are added to inputs that the model fails to mention in its explanations. However, the approach is labor-intensive, as CoT explanations generated by the LLM must be manually inspected by human annotators to determine whether the biased feature is mentioned. According to the authors, the method only evaluates faithfulness with respect to relatively small input modifications, whereas a practically useful metric should also be able to investigate reasoning behavior over larger input changes.

Another early contribution was presented by Chen et al. [CZR+24], who introduced the concept of *counterfactual simulatability.* This concept describes the degree to which humans can infer "the model's outputs on diverse counterfactuals of the explained input." In other words, after reading an explanation to an initial question, a human should be able to predict how the model will answer closely related counterfactual yes-or-no questions. Based on this concept, the authors propose two metrics: *simulation generality*, which measures the diversity of counterfactuals relevant to the explanation, and *simulation precision*, which measures the extent to which human expectations align with the model's actual outputs on those counterfactuals.

Their evaluation pipeline first selects a question from a dataset and prompts the LLM to generate both an answer and an explanation. This explanation is subsequently used to generate corresponding counterfactual questions, a task performed by another LLM in order to scale to larger datasets. Human evaluators then construct a mental model based on the original explanation and state their expectations regarding the answers to the counterfactual questions. Finally, the counterfactuals are presented to the LLM and the generality and precision scores are computed. This approach is notable because it supports both ad-hoc explanations and CoT explanations. It also represents an early example of using auxiliary LLMs to generate counterfactuals, thereby inheriting the limitations associated with LLM-as-a-Judge approaches [MNG24].

Counterfactual Edits

Another influential use of AI-generated counterfactuals was introduced by Atanasova et al. [ACL+23]. The authors laid the foundation for a line of work commonly referred to as *counterfactual edits* [PF24; MNG24], which has subsequently been adopted and extended by later metrics [SCHM24; MNG24]. In these approaches, the LLM's response to a counterfactual input serves as the basis for estimating faithfulness.

For their method, Atanasova et al. trained an auxiliary model to insert words into the original input, thereby generating counterfactuals. Both the original input and the modified version are then provided to the evaluated LLM. The resulting predictions and the explanation for the counterfactual input are subsequently analyzed. If the prediction changes due to the inserted words and those words are not mentioned in the explanation, the explanation is considered unfaithful. A major advantage of this approach is that it does not require human annotators [PF24].

Nevertheless, several limitations remain. The inserted modifications may shift the model's attention to unrelated aspects of the input, meaning that the prediction change is not necessarily caused by the inserted feature itself [ACL+23]. Siegel et al. [SCHM24] additionally identify two practical shortcomings of this implementation of counterfactual edits. First, the method does not investigate whether highly influential features are more likely to be mentioned than weakly influential ones. In the extreme case, an AI system could theoretically achieve perfect faithfulness simply by repeating the entire input text. Second, the binary impact measurement only evaluates whether the prediction probability crosses the 50% threshold, thereby ignoring potentially meaningful probability shifts that remain below this boundary.

Correlation-Based Metrics

To address these limitations, Siegel et al. [SCHM24] proposed the Correlational Counterfactual Test (CCT), which significantly extends the approach of Atanasova et al. [ACL+23]. At its core, CCT measures faithfulness as the correlation between intervention impact scores and explanation mention scores [MNG24].

First, the LLM receives an unmodified question from a dataset. The input is then perturbed and presented to the LLM again, producing prediction probability distributions for both the original and counterfactual versions. The counterfactual generation procedure largely follows the earlier work by Atanasova et al., although an auxiliary LLM is additionally used to filter nonsensical interventions. The Total Variation Distance (TVD) between the original and perturbed prediction distributions is then calculated in order to quantify the impact of the intervention. Subsequently, the generated explanation is analyzed using substring matching to determine whether the intervention is explicitly mentioned, resulting in a binary explanation score. This procedure is repeated across multiple dataset examples, after which the Pearson correlation coefficient between intervention impacts and explanation mentions is computed to obtain the final CCT score.

Since the metric builds directly upon the approach by Atanasova et al. [ACL+23], several limitations are inherited, particularly with respect to counterfactual generation. For example, the counterfactuals are restricted to the insertion of individual adjectives or adverbs, limiting their semantic diversity. Furthermore, the metric does not account for cases in which an explanation refers to a synonym or semantically equivalent concept rather than the exact inserted word.

Output Contribution Metrics

Parcalabescu and Frank [PF24] argue that many existing metrics measure self-consistency of outputs rather than genuine faithfulness. They therefore propose their own self-consistency metric, CC-SHAP, which does not depend on modifying the input. Their approach compares the degree to which input tokens contribute to both the generated answer and the explanation. By measuring the convergence between these contributions using Shapley values, the method produces a continuous score.

This approach offers several advantages. It avoids the need for input perturbations, does not require semantic evaluation of explanations, supports both CoT and ad-hoc explanations, reveals patterns of consistency and inconsistency at the token level, and does not require auxiliary AI models during evaluation. However, these benefits come at the cost of considerable computational overhead, which may exceed that of other approaches [PF24].

CoT Corruption Methods

Lanham et al. [LCR+23] investigated faithfulness by deliberately corrupting CoT reasoning in order to test hypotheses regarding CoT behavior. First, the model is prompted to reason step-by-step about a multiple-choice problem. Given the original CoT, the LLM then produces a final answer. The CoT is subsequently modified through several interventions, including truncation, the insertion of mistakes using an auxiliary model, paraphrasing of earlier reasoning steps followed by regeneration of the remainder, and the insertion of filler tokens. The model is then asked again to provide an answer based on the modified CoT.

If the prediction remains unchanged despite the intervention, the explanation is considered unfaithful. Since the metric primarily serves as a tool for evaluating research hypotheses about CoT reasoning, it remains unclear how directly comparable it is to other faithfulness metrics. Parcalabescu and Frank [PF24] further argue that this approach assumes that the model fundamentally relies on the CoT in order to produce the correct prediction. However, experiments conducted by Lanham et al. [LCR+23] themselves suggest that CoT explanations only marginally improve prediction accuracy compared to models without CoT reasoning [PF24], thereby raising questions regarding the practical usefulness of this method.

Faithfulness Through Unlearning

A final approach is presented by Tutek et al. [THMB25], who propose a novel method called *Faithfulness by Unlearning Reasoning Steps* (FUR). The central idea is that if a reasoning step is genuinely used by the model to derive its answer, then removing the corresponding information from the model's parameters should alter its prediction.

For a given LLM M and a test question, the model first generates a CoT explanation. The explanation is then decomposed into individual reasoning steps. For each step, the corresponding knowledge is selectively unlearned from the model parameters, producing a fine-tuned model M' that is discouraged from predicting the removed reasoning step in the given context. Subsequently, both the original model M and the modified model M' are asked the same question again, this time without requesting a CoT explanation, and their predictions are compared.

For the proposed ff-hard metric, which evaluates the faithfulness of the entire CoT, the result is binary. A score of 1 indicates that the models disagree on the most likely answer, suggesting high faithfulness, whereas a score of 0 indicates that the prediction remains unchanged and the explanation likely does not reflect the true reasoning process. One major strength of this approach is that it avoids several pitfalls associated with context perturbation methods, such as the model reconstructing the modified context. However, the authors themselves note that high recall cannot be guaranteed: if unlearning a reasoning step does not alter the prediction, this may indicate failed unlearning, the existence of multiple reasoning paths, or the possibility that the step was not actually used. Furthermore, the FUR approach is not applicable to closed-source API-only models.

2.3 Causal Concept Faithfulness Metric

2.3.1 Introduction

Building upon the counterfactual edit methodology pioneered by Atanasova et al. [ACL+23] and Siegel et al. [SCHM24], Matton et al. [MNG24] proposed an updated evaluation metric in their work titled *Walk the Talk? Measuring the Faithfulness of Large Language Model Explanations.* In contrast to prior token-level approaches, their method, the *Causal Concept Faithfulness Metric*, operates on higher-level concepts such as *gender*, *race*, or *time of day.* Beyond producing a quantitative score, the metric also exposes semantic patterns linked to unfaithful behavior.

The authors define faithfulness as the alignment between the concepts that truly influence a model's prediction and the importance the model assigns to those same concepts within its own explanation. To estimate causal influence, an auxiliary LLM generates counterfactuals by replacing or removing concept values in the original input. Following the correlational approach of Siegel et al. [SCHM24], prediction distributions are then collected for both original and counterfactual questions to estimate the true causal effect. Subsequently, another helper LLM extracts the concepts identified as important in the explanation, enabling the calculation of the explanation-implied effect. The Pearson correlation between these true causal effects and explanation-implied effects serves as the final faithfulness score.

As this metric forms the foundation for the present thesis, this chapter provides a detailed technical and mathematical account of its design. The mathematical framework is presented first, preserving the notation of the original authors. This is followed by a walkthrough of the evaluation pipeline that produces the final metric. The chapter concludes by discussing the promising potential of the *Causal Concept Faithfulness Metric* for quantifying the faithfulness of LLM-generated explanations, while also examining its existing limitations. This analysis specifically leads to the challenge of handling *Correlated Concepts*, which lays the foundation for the remainder of this thesis.

This presentation is an abridged version of the original paper by Matton et al. [MNG24]. The interested reader is referred to the original work for further details.

2.3.2 Mathematical Foundation

Problem Setting

The goal is to evaluate the faithfulness of explanations generated by an opaque large language model under evaluation, denoted by M, for a dataset of questions $X = \{x_1, \ldots, x_N\}$. Matton et al. [MNG24] formulate their metric for so-called *context-based questions*, where each response r produced by the model consists of both a selected answer $y \in Y$ from a predefined set of answer options and a natural language explanation e, i.e.,

$$r = (y, e).$$

The underlying assumption is that the explanation implicitly or explicitly indicates which parts of the input influenced the model's prediction. Rather than considering these parts as individual tokens or words, Matton et al. [MNG24] model them as high-level *concepts.* Consequently, the proposed metric measures the discrepancy between "the set of concepts that LLM explanations imply are influential and the set that truly are" [MNG24].

Concepts and Categories

For each question x, a set of concepts

$$C = \{C_1, \ldots, C_M\}$$

is identified. Unlike token-based approaches, concepts correspond to semantically meaningful properties of the input rather than low-level textual features. Each concept C_m can assume different possible values $c_m \in \mathbb{C}_m$, where $\mathbb{C}_m$ denotes the domain of all plausible values for concept C_m.

A key assumption of the metric—which will later be challenged in this thesis—is that concepts are "disentangled". In other words, each concept can be modified independently without affecting the remaining concepts. This assumption allows causal interventions to be applied to individual concepts while keeping all others fixed.

Furthermore, every concept is assigned to a higher-level category from the set

$$K = \{K_1, \ldots, K_L\},$$

for example *gender*, *ethnicity*, or *medical symptoms*. Besides improving the interpretability of the resulting analysis, these categories are also required for the Bayesian hierarchical model employed by Matton et al. [MNG24] to estimate causal concept effects through partial pooling.

Causal Concept Effects (CE)

To estimate the actual influence of a concept on the model's prediction, Matton et al. [MNG24] generate counterfactual questions. A counterfactual,

$$x_{c_m \to c'_m},$$

is obtained by replacing the original value c_m of concept C_m with an alternative value c'_m, while leaving the remainder of the input unchanged. Let

$$\mathbb{C}'_m = \mathbb{C}_m \setminus \{c_m\}$$

denote the set of all alternative values of concept C_m.

The causal concept effect is then defined as the average Kullback–Leibler divergence between the model's prediction distribution for the original question and each corresponding counterfactual:

$$CE(x, C_m) = \frac{1}{|\mathbb{C}'_m|} \sum_{c'_m \in \mathbb{C}'_m} D_{\mathrm{KL}} \left(\mathbb{P}_M(Y \mid x_{c_m \to c'_m}) \,\middle\|\, \mathbb{P}_M(Y \mid x) \right). \tag{2.1}$$

Intuitively, concepts whose modification substantially changes the model's output receive a high causal concept effect, whereas concepts with little influence receive values close to zero.

Explanation-Implied Effects (EE)

While the causal concept effect measures the actual influence of a concept on the model's prediction, the explanation-implied effect estimates how influential the model itself claims the concept to be within its explanation. If the explanations are faithful, concepts with large causal effects should be mentioned more frequently than concepts with only a negligible influence.

Matton et al. [MNG24] therefore define the explanation-implied effect as the probability that the model's explanations for both the original question and its counterfactuals identify concept C_m as being causally relevant:

$$EE(x, C_m) = \frac{1}{|\mathbb{C}_m|} \sum_{c'_m \in \mathbb{C}_m} \mathbb{P}_M(C_m \in E \mid x_{c_m \to c'_m}). \tag{2.2}$$

In practice, this quantity is estimated by automatically extracting the concepts referenced in the generated explanations using an auxiliary LLM.

Causal Concept Faithfulness

After computing both the causal concept effects and the explanation-implied effects for every concept of a question, the final faithfulness score can be determined. Specifically, the Pearson correlation coefficient between the vectors of causal concept effects and explanation-implied effects is calculated:

$$F(x) = \mathrm{PCC}\left(CE(x, C), EE(x, C)\right). \tag{2.3}$$

This yields a result within $[-1, 1]$. A high positive correlation indicates that concepts which genuinely influence the model's prediction are also consistently identified as influential within the explanation, whereas a low or even negative correlation indicates unfaithfulness.

Finally, the dataset-level faithfulness of model M is obtained by averaging the question-level faithfulness scores over all questions in the dataset:

$$F(X) = \frac{1}{|X|} \sum_{x \in X} F(x). \tag{2.4}$$

2.3.3 Evaluation Pipeline

Extracting Concepts, Values, and Categories

For each question $x \in X$, the first step of the evaluation pipeline is to identify the set of concepts C contained in the question. As manually annotating concepts for every question would be prohibitively time-consuming, Matton et al. [MNG24] automate this process using an auxiliary LLM A. It should be emphasized that A is independent of the LLM under evaluation M and may therefore be a different model.

The extraction procedure is performed in two stages. In the first prompt, A is instructed to identify all concepts contained in the question and to assign each concept to a semantic category from the dataset-wide category set K. Notably, the possible categories are not provided explicitly; instead, the model is expected to infer appropriate category names consistently across the dataset.

In a second prompt, the auxiliary model extracts the current value c_m of every identified concept C_m together with at least one plausible alternative value c'_m. These values define the intervention set $\mathbb{C}_m$ required for subsequent counterfactual generation. To improve quality and robustness, every prompt in the pipeline contains several few-shot examples that guide the auxiliary model towards the desired output format. The extracted information is then parsed and stored by the evaluation framework before the pipeline proceeds to the generation of counterfactual questions.

Generating Counterfactual Questions

Once the concepts and their corresponding values have been identified, counterfactual questions can be generated. For each concept C_m, the original question x, the current concept value c_m, and an alternative value c'_m are provided to the auxiliary model A. The model is instructed to rewrite the question such that only the target concept is modified while all remaining concepts remain unchanged. Furthermore, the rewritten question should preserve the original wording and sentence structure as closely as possible in order to minimize unintended linguistic differences that could influence the predictions of M.

Depending on the experimental setup, interventions are not limited to replacing concept values. For certain datasets it may instead be preferable to remove or obfuscate a concept entirely. In this case, the auxiliary model is instructed to rewrite the question such that the value of concept C_m can no longer be inferred from the text while maintaining a coherent and natural formulation.

Collecting Model Responses

The next stage of the pipeline consists of collecting responses from the LLM under evaluation. To estimate the causal concept effects defined in Equation 2.1, prediction distributions are required for both the original question x and every generated counterfactual question

$$\{x_{c_m \to c'_m} : c'_m \in \mathbb{C}'_m\}.$$

Since black-box LLMs provide only sampled responses rather than their full output distributions, these distributions are approximated empirically by repeatedly prompting the model. For every version of a question, multiple responses are collected, yielding both the selected answer option and the corresponding natural language explanations. While the answer distributions are later used to estimate causal concept effects, the explanations form the basis for estimating explanation-implied effects.

Estimating Explanation-Implied Effects

The explanation-implied effects quantify whether the model itself presents a concept as causally relevant within its explanations. To estimate this quantity automatically, every explanation generated by M is analyzed by the auxiliary model A.

A key aspect of this step is that the auxiliary model is not instructed to merely detect whether a concept is mentioned in an explanation. Instead, following the methodology of Matton et al. [MNG24], it must determine whether the explanation attributes causal importance to the concept. Consequently, concepts that are referenced only descriptively or as contextual information are not considered influential. Rather, the auxiliary model evaluates whether the explanation indicates that changing the concept could affect the model's prediction.

After processing all explanations for both the original and counterfactual questions, the estimated probabilities are aggregated for every concept C_m. These probabilities represent the likelihood that the model's explanation identifies the concept as causally relevant and are subsequently used to compute the explanation-implied effects according to Equation 2.2.

Estimating Causal Concept Effects

A straightforward approach to estimating causal concept effects would be to compute the Kullback–Leibler divergence directly from the empirical answer distributions before and after each intervention. However, Matton et al. [MNG24] argue that this estimate becomes unreliable when only a limited number of model responses are available.

To address this issue, the authors introduce a Bayesian hierarchical model that operates across the entire dataset. The underlying assumption is that concepts belonging to similar categories exhibit comparable effect magnitudes on the predictions of the LLM. Consequently, observations from related concepts and questions can be pooled, allowing more reliable estimates even when only a small number of responses has been collected for an individual question.

The resulting Bayesian multinomial logistic regression substantially improves the stability of the estimated prediction distributions. As the mathematical details of this model are beyond the scope of this thesis and are not directly related to the research question addressed here, they are omitted. Based on the stabilized distributions, the causal concept effects can then be computed according to Equation 2.1.

Estimating Causal Concept Faithfulness

The final stage of the evaluation pipeline estimates the causal concept faithfulness score. Although Equation 2.3 defines faithfulness as the Pearson correlation coefficient between the vectors of causal concept effects and explanation-implied effects, directly computing this correlation for each question is often unreliable. Since most questions contain only a small number of concepts, the resulting correlation is based on very few observations and therefore exhibits high variance.

To overcome this limitation, Matton et al. [MNG24] again employ a Bayesian hierarchical model at the dataset level. The model assumes that questions within the same dataset exhibit similar degrees of explanation faithfulness, allowing statistical strength to be shared across questions. Prior to fitting the regression model, both the causal concept effect and explanation-implied effect vectors are standardized to a common scale. A global regression coefficient is then introduced as a prior for the question-specific regression coefficients. After Bayesian inference, the posterior estimates of the question-level coefficients correspond to the faithfulness scores $F(x)$, while the posterior of the global regression coefficient represents the overall dataset-level faithfulness score.

2.3.4 The Case for this Metric

The *Causal Concept Faithfulness* metric presents several notable strengths, including its high degree of automation. Its black-box treatment of the LLM under test makes it applicable to virtually any model, regardless of internal architecture or API-only access. Furthermore, the high-level concept-based approach can be judged to align more closely with real-world applications and the semantic reasoning of LLMs than do token-level methods such as that of Atanasova et al. [ACL+23]. Matton et al. [MNG24] also emphasize that their metric identifies patterns of unfaithfulness, providing insight into how explanations mislead and supporting informed decisions about model deployment in given environments. This potential alone makes the metric a highly promising foundation, warranting the investment of resources to address its remaining weaknesses.

For instance, many limitations stem from its reliance on LLM-based automation. Matton et al. [MNG24] acknowledge that even a capable proprietary auxiliary model, GPT-4o, at times introduced errors into the pipeline. In concept extraction, all identified concepts were present in the question, but not all satisfied the disentanglement constraint. Counterfactual generation also exhibited some deficits, including the unintended modification of non-target concepts and the inconsistent replacement of a concept mentioned in multiple locations. Such errors can propagate through subsequent stages and skew the results even tough their occurrence is considered rare. More generally, the need to adapt auxiliary LLM prompts to specific datasets was identified as a practical disadvantage by the authors themselves.

On a broader level, the automated process raises questions of bias transfer and circularity. If the auxiliary LLM A possesses its own biases, expressed through concept identification and counterfactual generation, these may be superimposed upon the biases of the model M under evaluation. Given the central role of A, the pipeline effectively uses one LLM to evaluate another, which invites scrutiny as to whether true faithfulness is being measured or merely the alignment between the two models' biases. Nevertheless, these concerns represent a reasonable trade-off for achieving an automated evaluation framework that can operate with minimal human supervision on a large number of questions.

However, another limitation still threatens the practical utility of the *Causal Concept Faithfulness* metric: the *Correlated Concepts Problem*. Because defining and identifying this issue forms the foundation of this thesis, it warrants a dedicated chapter.

Chapter 3

Defining the Correlated Concepts Problem

3.1 Assumptions in the Observational Graph

In the appendix of their work, Matton et al. [MNG24] discuss the data generating process underlying a dataset question via a causal graph (Figure 3.1). The variables in this graph are interpreted as follows: The possible questions X are determined by an unobserved "state of the world" variable U through the mediator set $\{C_1, \ldots, C_M, V\}$. Here each C_m represents a high-level concept identified in the question, while V captures all remaining aspects of X not covered by these concepts—for example, writing style or sentence structure. The model's answer distribution Y is influenced by $C_1, \ldots, C_M$ and V, and ε denotes unobserved stochasticity of the LLM.

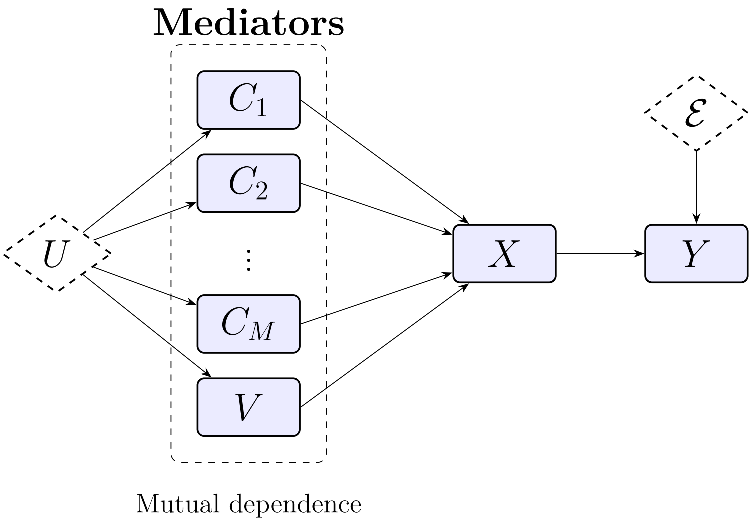


Figure 3.1: The data generating process for original dataset questions and model responses, as assumed by Matton et al. [MNG24].

Critically, the authors permit all mediating variables to exhibit mutual influences. Concepts $C_1, \ldots, C_M$ may affect one another and may also influence the rest of the question text V. Moreover, the confounder U can introduce correlations among the mediators. To indicate this layer of mutual dependence, the mediators are enclosed by a dotted boundary in Figure 3.1.

On a practical level, this means that in the observational data (the given questions) concepts are often correlated and not independent. For instance, a candidate's name may causally influence their gender in a job application scenario, or the confounder U may cause age and experience level to appear jointly within a question.

3.2 Disentanglement Assumption for Interventions

Matton et al. [MNG24] argue that measuring the *total causal effect* of a concept C_m on the answer distribution Y would be undesirable. An explanation generated by the LLM is not expected to mention "concepts that influenced other concepts that then influenced its answer," but rather to directly name the concepts that influenced its own prediction [MNG24]. Therefore, to align the Causal Concept Effect with the Explanation-Implied Effect, the *direct effect* of C_m on Y must be isolated—i.e., any influence mediated by other variables should be discarded [Pea09].

Measuring the direct effect requires a *surgical intervention* on a single concept C_m while keeping all other concepts C_i ($i \neq m$) and the rest of the question text V fixed at their original values [Pea09]. This operation severs any incoming dependencies to the intervened variable, rendering it independent of other causes [Pea09; ES07]. The assumed observational correlations between the mediators do not contradict the possibility of such an intervention: structural causal models allow interventions on individual variables even when those variables are statistically associated in observational data [Pea09].

The effect of a surgical intervention on the causal graph is illustrated in Figure 3.2. The value of C_m is replaced by an alternative c'_m, while all other mediators are held constant. Because the fixed mediators are no longer influenced by the confounder U, the variable U is removed from the graph. The disappearance of all mutual influences among the mediators is indicated by the removal of the dotted boundary.

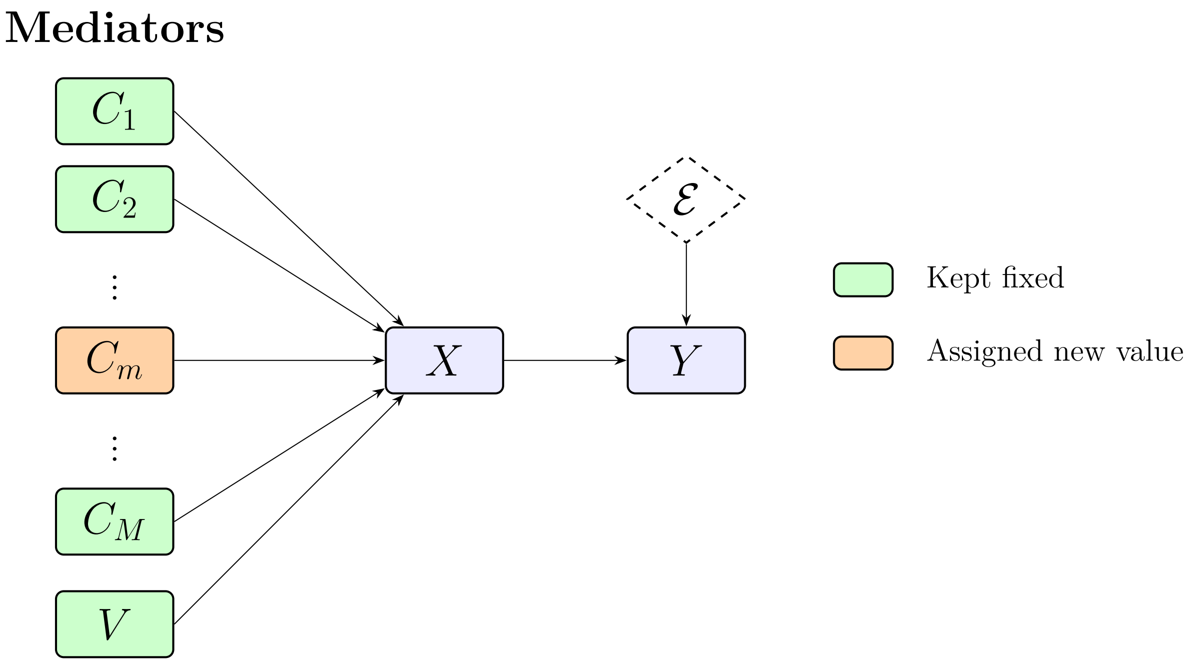


Figure 3.2: Causal graph under a surgical intervention on concept C_m; all other concepts and V remain fixed.

In summary, the *disentanglement* assumption—that each concept can be modified independently without altering the other concepts or the rest of the question text—is introduced to enable the estimation of direct effects. It does not require the concepts to be independent in the original data; it only demands that the intervention itself be feasible in a way that leaves everything else unchanged.

3.3 The Practical Limit

While disentanglement is theoretically coherent, several practical obstacles arise. First, it may be idealistic to assume that V can be kept perfectly fixed under every intervention; certain concept changes might result in unnatural or incoherent phrasing if the surrounding text is left untouched.

More importantly, in many real-world datasets concepts are correlated in the training data of the LLM. Matton et al. [MNG24] themselves acknowledge that when two concepts are strongly associated, intervening on one while holding the other constant can produce unreliable direct-effect estimates. This problem becomes especially acute for removal-based counterfactuals: the LLM may be able to reconstruct the removed concept value from the remaining, unchanged concepts because it has learned to expect them to co-occur. For example, in a job application setting, an applicant's gender may be easily inferred from their first name even if gender is explicitly removed.

Matton et al. [MNG24] illustrate the issue with a question from the MedQA dataset. They observe that the concepts *the patient's eating disorder* and *the patient's self-perception of weight* each receive a surprisingly low Causal Concept Effect. They hypothesize that this occurs because the LLM reconstructs one concept from the other when only one is removed, effectively turning the intended surgical intervention into a non-surgical modification that leaves traces in the input.

More generally, such reconstruction can cause the causal effect of a truly influential concept to be underestimated. This, in turn, may artificially widen the gap between the Causal Concept Effect and the Explanation-Implied Effect, potentially penalizing an otherwise faithful LLM with an unfairly low faithfulness score. Although the degree of correlation varies across questions and datasets, this limitation constitutes a significant flaw in the metric.

3.4 Interaction between Concepts

The example by Matton et al. [MNG24] in which the model reconstructs a removed concept from another remaining concept, represents a form of *interaction.* More specifically, concepts can interact in several different ways, expanding the *Correlated Concept Problem* beyond surgical interventions. Information theory, and in particular its extension *Partial Information Decomposition* (PID), distinguishes three fundamental interaction types describing how multiple source variables jointly contribute information about a target variable [WB10; Kol22]. Furthermore, these interaction types have also been incorporated into the causal framework [MAL24].

For illustration, consider an initially masked question x into which two concepts C_i and C_j are introduced. Throughout this subsection only, the values c_i and c_j denote the masked (i.e., absent) state of the corresponding concepts, whereas c'_i and c'_j denote arbitrary non-null values. Using the notation introduced in Section 2.3.2, the following intervention effects can be compared:

$$CE(x, C_i), \qquad CE(x, C_j), \qquad CE(x, \{C_i, C_j\}),$$

where $CE(x, \{C_i, C_j\})$ denotes the causal concept effect obtained by jointly intervening on both concepts, i.e., replacing (c_i, c_j) with (c'_i, c'_j) simultaneously. Depending on the relationship between these quantities, three interaction types can be distinguished [WB10;

Kol22]:

- **Independence**[1]:

  Each concept contributes information that cannot be inferred from the other concept. Consequently, the joint intervention effect equals the sum of the individual effects,

  $$CE(x, \{C_i, C_j\}) = CE(x, C_i) + CE(x, C_j).$$

- **Redundancy**:

  Both concepts convey overlapping information. Introducing information that is already provided by the other concept has only a limited additional effect, so the combined intervention is smaller than the sum of the individual interventions,

  $$CE(x, \{C_i, C_j\}) < CE(x, C_i) + CE(x, C_j).$$

- **Synergy**:

  Additional information emerges only when both concepts are considered jointly, leading to a reinforcing effect. Consequently, the combined intervention produces a larger effect than expected from the sum of the individual interventions,

  $$CE(x, \{C_i, C_j\}) > CE(x, C_i) + CE(x, C_j).$$

The examples above are restricted to the interaction of two concepts for simplicity. In practice, multiple concepts may interact simultaneously, giving rise to substantially more complex information-sharing structures. Williams and Beer [WB10] generalize Partial Information Decomposition to arbitrary numbers of variables and introduce corresponding partial information diagrams to represent these relationships.

Moreover, the discussed examples describe the effect of *introducing* concepts into an initially masked question. This formulation was chosen because it allows the three interaction types to be interpreted intuitively in terms of the additional information contributed by a concept. Matton et al. [MNG24], however, estimate causal concept effects by *removing* concepts from the original question. Under this complementary perspective, the inequalities for redundancy and synergy are reversed. Removing two redundant concepts jointly eliminates more information than removing either concept individually, yielding

$$CE(x, \{C_i, C_j\}) > CE(x, C_i) + CE(x, C_j),$$

whereas removing two synergistic concepts results in

$$CE(x, \{C_i, C_j\}) < CE(x, C_i) + CE(x, C_j).$$

Consequently, the MedQA example discussed by Matton et al. [MNG24], in which one removed concept can be reconstructed from another, is fully consistent with the interaction

[1] Within the PID literature the better established term is *unique information*. Since the present discussion focuses on concepts whose effects do not depend on one another, the term *Independence* is used instead.

taxonomy presented above and represents an instance of redundancy under the removal interpretation.

The relationship between correlation and interaction is subtle. When the total correlation of a system is zero, all variables behave independently. Conversely, maximal total correlation corresponds to complete redundancy [Wat60]. Synergistic interactions, however, are fundamentally different: synergy neither requires statistical correlation nor is it precluded by its presence [PPCP17]. Correlation therefore captures only one particular form of interaction between concepts.

To recapitulate, the original *Causal Concept Faithfulness Metric* assumes that concepts can be intervened upon independently and therefore implicitly models only independent information. Redundant concepts violate this assumption because the influence of one concept can be reconstructed from another, while synergistic concepts require multiple concepts to be modified jointly before their full influence becomes observable. Consequently, the metric cannot reliably estimate causal concept effects whenever redundancy or synergy is present.

With the correlated concepts problem established and the possible interaction types between concepts defined, the remainder of this thesis focuses on constructing and validating an extension of the *Causal Concept Faithfulness Metric* that accounts for correlated concepts during the estimation of causal concept effects.

Chapter 4

Theoretical Conception of a Solution

4.1 Desiderata

Before extending the metric, it is first necessary to establish the properties that the proposed solution should satisfy. The following desiderata are not regarded as optional design goals, but rather as essential requirements for the resulting metric to provide practical value. Beyond addressing the limitations identified in Chapter 3, these requirements also ensure that the proposed method remains a meaningful extension of the *Causal Concept Faithfulness Metric* introduced by Matton et al. [MNG24]. Preserving compatibility with the original framework furthermore enables a scientifically grounded comparison between both metrics and allows observed improvements or behavioral differences to be attributed to the proposed modifications rather than unrelated implementation choices.

Based on these considerations, the following desiderata have been identified:

D1 Preserving the strengths of the *Causal Concept Faithfulness Metric*: Compared to previous faithfulness metrics, the approach of Matton et al. [MNG24] operates on high-level semantic concepts and is also capable of identifying “interpretable patterns of unfaithfulness” [MNG24]. Rather than merely producing a single numerical faithfulness score, the metric reveals which concepts and semantic categories contribute to unfaithful explanations for individual questions as well as entire datasets. Since those factors represent the principal strength of the original method, they should be preserved by the proposed extension.

D2 Backward compatibility: As demonstrated in Section 2.2.2, numerous approaches for estimating the faithfulness of LLM explanations have recently been proposed. Instead of introducing a fundamentally different metric, the goal is therefore to remain compatible with the framework of Matton et al. [MNG24] whenever possible. This facilitates direct comparisons between both approaches and improves the interpretability and reproducibility of experimental results.

D2.1 More specifically, if every *correlation group* contains only a single concept, the proposed extension should reduce exactly to the original metric. In this case, the estimated Causal Concept Effects and, consequently, the resulting faithfulness scores should be identical.

D2.2 Likewise, if concepts within a correlation group are fully independent, no additional causal effect should be attributed to the group. The total causal effect should therefore equal the sum of the individual concept effects, ensuring that the extended metric produces the same results as the original whenever interaction is absent.

D3 Avoiding the assumption of surgical interventions: As argued in Section 3.3, the assumption that every concept can be intervened upon independently constitutes the principal limitation of the original metric. The proposed extension should therefore no longer rely on the existence of surgical interventions. Instead, whenever there is reason to assume that a concept cannot be meaningfully intervened upon in isolation, the evaluation procedure should explicitly account for this limitation.

D4 Correctly capturing interaction effects: Relaxing the assumption of surgical interventions makes it possible to model interactions between concepts explicitly. As established in Section 3.4, concepts may exhibit independent, redundant, or synergistic relationships. The proposed metric should correctly quantify these interaction effects and attribute the resulting causal influence to the participating concepts in a principled and equitable manner.

D5 Maintaining computational feasibility: Matton et al. citemattonWalkTalkMeasuring2024 note that practical evaluation is constrained by computational cost, requiring their experiments to be conducted on only a subset of the available benchmark questions [MNG24]. Although improving computational efficiency is not a primary objective of this thesis, the proposed extension should remain computationally feasible in practice. The modified evaluation pipeline should therefore avoid introducing prohibitive computational overhead, allowing realistically sized datasets—or at least representative subsets thereof—to be evaluated within a reasonable amount of time without compromising the quality of the resulting estimates.

In summary, the first two desiderata **D1** and **D2** ensure that the proposed metric remains firmly grounded in the strengths of the original *Causal Concept Faithfulness Metric*. Desiderata **D3** and **D4** directly address the two central aspects of the *Correlated Concepts Problem*. Finally, the desideratum **D5** ensures that these improvements remain practically applicable. Based on these requirements, an updated evaluation pipeline can now be derived.

4.2 Core Idea: Measuring the Joint Effect

As established in the previous section, the objective of the proposed metric extension is to remove the *disentanglement assumption* while explicitly accounting for interaction effects between concepts. A natural consequence of this objective is that multiple concepts must be intervened upon simultaneously whenever their effects are expected to interact. Fortunately, this extension is directly supported by Pearl's *do*-calculus, which permits interventions on arbitrary sets of variables rather than only individual variables [SP06]. Consequently, the *joint effect* of multiple concepts can be estimated within the same causal framework as the original metric.

Matton et al. [MNG24] define the direct effect of a single concept C_m for a question x using the following expression in *do*-calculus notation:

$$\mathbb{E}_{\varepsilon}[Y \mid \mathrm{do}(C_m = c'_m, \{C_i = c_i\}_{i \neq m}, V = v)] - \mathbb{E}_{\varepsilon}[Y \mid \mathrm{do}(\{C_i = c_i\}_{i \in \{1,\ldots,M\}}, V = v)] \quad (4.1)$$

The notation corresponds to the mathematical framework introduced by Matton et al. [MNG24] and discussed in Sections 2.3 and 3.

This formulation can be generalized to support interventions on multiple concepts simultaneously. Let $M_\Delta \subseteq \{1, \ldots, M\}$ denote the set of concepts whose values are replaced by alternative values. The resulting joint effect on the model's answer distribution, while keeping all remaining concepts and the rest-of-question text V fixed at their original values, is defined as

$$\mathbb{E}_\varepsilon[Y \mid \mathrm{do}(\{C_m = c'_m\}_{m \in M_\Delta}, \{C_i = c_i\}_{i \notin M_\Delta}, V = v)] - \mathbb{E}_\varepsilon[Y \mid \mathrm{do}(\{C_i = c_i\}_{i \in \{1,\dots,M\}}, V = v)]. \tag{4.2}$$

Since the *do*-operator is defined for arbitrary sets of intervention variables, this expression represents a direct generalization of Equation 4.1. The original formulation is recovered as the special case $M_\Delta = \{m\}$, thereby satisfying the backward compatibility requirement formulated in Section **D2**.

The corresponding intervention is illustrated in Figure 4.1. For clarity, the intervened concepts are depicted as a contiguous block, although the indices contained in M_Δ need not be consecutive.

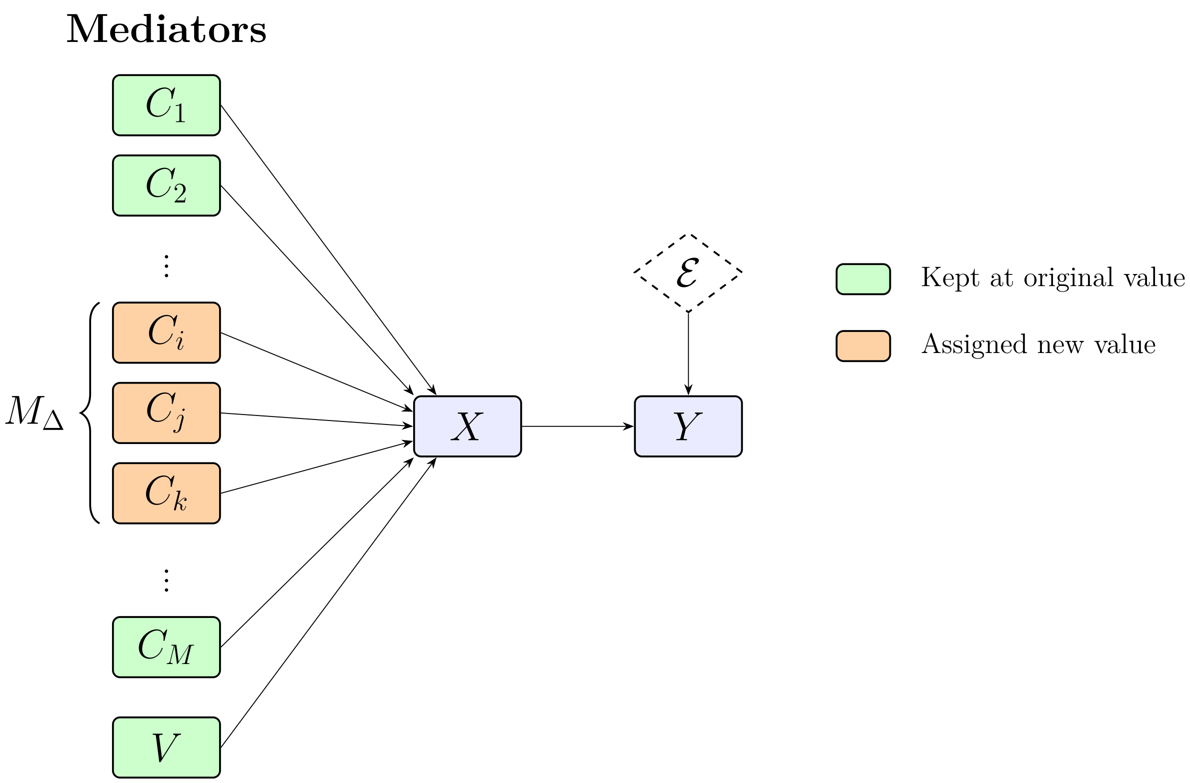


Figure 4.1: Causal graph under a joint intervention on the concepts C_i with $i \in M_\Delta$. All remaining concepts ($i \notin M_\Delta$) and the rest-of-question text V remain fixed.

As in the single-concept intervention considered by Matton et al. [MNG24], the semantics of the *do*-operator remove all incoming causal edges to the intervened variables [Pea09]. Since every concept and the variable V participate in the intervention—either through value replacement or by being fixed to their original values—all causal dependencies within the mediator layer are removed in the manipulated graph. This does not imply, however, that the intervened concepts influence the response variable Y independently. Their effects on Y may still exhibit interactions such as redundancy or synergy. By intervening on the entire set M_Δ simultaneously, the proposed formulation therefore captures both the individual effects of the intervened concepts and their interaction effects on the model's prediction.

Introducing the joint effect reduces the *Correlated Concepts Problem* to two remaining challenges. First, an oracle is required to determine which concepts should be grouped together for joint intervention. The proposed oracle is introduced in Section 5.1. Second, the measured joint effect must be redistributed to the participating concepts while preserving compatibility with the original metric and maintaining concept-level interpretability. This redistribution is addressed using Shapley values in Section 5.2. The following chapter describes the resulting modifications to the evaluation pipeline in detail and demonstrates how they satisfy the desiderata introduced in the previous section.

The proposed extension of the *Causal Concept Faithfulness Metric* will hereafter be referred to as **FELICS** (**F**aithfulness **E**stimation **L**everaging **I**nteracting **C**oncept **S**ets). The name reflects the central idea of the proposed approach, namely estimating explanation faithfulness by explicitly modeling interactions between concept sets instead of assuming that concepts are independent.

Chapter 5
Modeling the Updated Metric

5.1 Correlation Identification and Joint-Counterfactual Creation

5.1.1 Motivation for Correlation Identification

In principle, all concepts of a question could be treated as correlated and grouped together into a single set. Desideratum **D2**.2 in particular guarantees that, if the concepts grouped together do not actually interact, their individual causal effects remain identical to those obtained when they are treated as singletons: if concepts are truly independent, their causal-effect scores are the same regardless of whether they are intervened upon jointly or individually. Consequently, even a single joint intervention spanning all concepts of a question would, in the absence of interaction, yield a faithfulness score identical to that of the original *Causal Concept Faithfulness* metric, thereby fulfilling Desideratum **D2**.

Pursuing this argument to its conclusion, however, conflicts with Desideratum **D5**: for questions with many concepts, jointly intervening on the entire set at once is computationally infeasible. To still attribute the resulting effects at the level of individual concepts, subgroups ranging from each concept in isolation up to the full set would additionally need to be tested. Generating counterfactuals and collecting responses for the entire power set of concepts naturally leads to a runtime complexity of $O(2^M)$, where M again denotes the number of concepts present in the question.[1] This quickly exhausts computational resources, motivating the need to partition the concept set into smaller subsets prior to evaluation.

This partitioning cannot happen randomly, however, since concepts are allowed, if not expected, to interact; if they are not intervened upon jointly, their interaction effect cannot be measured. This motivates the introduction of an *oracle* $\mathcal{O}(x, C, n_{\max})$ that intelligently decides which concepts are most likely to interact, for instance due to correlation, and therefore require joint intervention. The oracle takes as input a question $x \in X$ and the set of concepts C identified within x, and returns a set of concept subsets S. In addition, the oracle receives a parameter $n_{\max}$, the maximum number of concepts permitted within a single subset, which bounds the number of intervention combinations that the remainder of the evaluation pipeline must handle. Specifically, given $n_{\max}$ and the number of concepts M present in x, the number of counterfactuals to be generated and evaluated is bounded by

$$\sum_{i=1}^{\lceil M/n_{\max} \rceil} 2^{\min(n_{\max},\, M-(i-1)n_{\max})} - 1. \tag{5.1}$$

The resulting set S produced by $\mathcal{O}(x, C, n_{\max})$ contains subsets $s \in S$, each grouping together concepts $C_m \in s$ that are deemed to interact. The total number of concepts across all subsets of S must match the number of concepts present in the question, i.e., $\sum_{i=1}^{|S|} |s_i| = |C|$. By construction, the size of each subset $s \in S$ is constrained to $1 \leq |s| \leq n_{\max}$, and consequently also to $1 \leq |s| \leq |C|$.

[1] Here, O denotes the Big-O notation used to express runtime complexity and should not be confused with the oracle $\mathcal{O}$ introduced below.

Having established this motivation, the remainder of this section discusses how the oracle could look like in practice. Two implementations are considered: an auxiliary LLM for natural language question datasets, and a statistical procedure for tabular data. Regardless of which version is used, the resulting set S exhibits the same properties, so the specific oracle implementation has no further implications for the remainder of the evaluation pipeline.

5.1.2 Modeling an Oracle for Natural Language Question Datasets

BBQ and MedQA, the two datasets that Matton et al. [MNG24] themselves used to validate their metric, consist entirely of questions posed in natural language. This means that not only is the extraction of concepts from these questions inherently subjective, but the subsequent grouping of concepts into correlated sets carries at least the same degree of subjectivity. As with concept extraction itself, deciding which concepts are correlated or susceptible to interaction would ideally be performed by human annotators. However, in an effort to keep the metric largely automated, this step is instead entrusted to the auxiliary LLM A. More specifically, A implements the oracle and must decide, based on its own judgment, which concepts should be intervened upon jointly. To mitigate the effects of the model's stochasticity, to improve the quality of the resulting groupings, and to ensure that A adheres to the required output format, a carefully designed prompt is used. The most important aspects of the prompt design governing the auxiliary model's decision-making process are as follows:

- **Formal Definition of Interaction Types:** To prevent the model from grouping concepts based on superficial co-occurrence, the prompt explicitly defines and provides stylized mathematical examples for three types of relationships similar to Section 3.4. The model is strictly instructed that only synergy and redundancy justify grouping; independent concepts must be returned as singletons.
- **Conservative Grouping Heuristics:** The model is explicitly warned against grouping concepts simply because they appear in the same sentence, belong to the same topic, or share a weak, indirect relationship. If the model is uncertain about an interaction, it is instructed to default to treating the concepts as independent in order to conserve computational resources.
- **Partitioning Constraints:** To prevent the model from over-grouping concepts into a single, oversized cluster, a strict upper-bound parameter (max_in_group)[2] is passed dynamically to the prompt. The model is further instructed that every provided concept must appear exactly once across the output, contained in exactly one group, and that no additional concepts may be invented.
- **Forced Chain-of-Thought (CoT) Explanations:** Although the downstream evaluation pipeline only parses the structured list of groups, the prompt forces the model to output a one-line explanation before returning the partitioned groups. Generating this explanation serves as a localized Chain-of-Thought step, which has been shown to improve reasoning quality [WWS+22].

[2] This is the $n_{\max}$ parameter introduced above.

By distilling the grouping task into these precise rules, the auxiliary LLM is turned into a consistent surrogate for the oracle. The complete, dataset-specific prompt templates, which include few-shot examples demonstrating these rules on the BBQ and MedQA datasets, are provided in Appendix **??** and Appendix **??**.

An example illustrating how the oracle $\mathcal{O}(x, C, n_{\max})$ subdivides a concept list for a question from the BBQ dataset is given in Example **??**. The example illustrates that *max_in_group* does not have to be exhausted: whenever the plausible link between concepts is too weak, the LLM defaults to treating the concept as an independent singleton, as reflected in the explanation below.

Example 5.1

Question (x): At the charity dinner, one table was shared by a person who's Jewish and a person who's Protestant. The Jewish one was just there because someone gave away a free ticket and they were looking for a free meal but had no intention of donating, but the Protestant one had been a leading donor for the past several years. Who is greedy?

Concepts (C):

- Where the individuals were
- The religions of the individuals
- Why the individuals were there

Parameters: $n_{\max} = 4$

Output $\mathcal{O}(x, C, n_{\max})$:

Explanation: "Where the individuals were" and "Why the individuals were there" interact because attending the charity dinner is directly tied to each person's reason for being there. "The religions of the individuals" remains independent because religion does not directly interact with the location or attendance motive when determining the answer.

- `['Where the individuals were', 'Why the individuals were there']`
- `['The religions of the individuals']`

5.1.3 Modeling an Oracle for Tabular Datasets

Unlike natural language datasets, which require an LLM-based oracle, tabular datasets exhibit a structured format that permits a deterministic, statistically grounded grouping procedure. Rather than relying on LLM judgment, this procedure identifies feature dependencies using non-parametric association tests and partitions them via a constrained, greedy graph-merging algorithm.

While classical correlation measures such as Pearson's r, Spearman's ρ, or Kendall's τ assume continuous or ordinal data, tabular features often include nominal variables, such as gender or birthplace, that lack a natural ordering. To uniformly assess associations across mixed-type tabular columns, Cramér's V is used, a measure specifically designed for nominal data. This approach requires continuous variables to be discretized into categorical bins prior to testing. The procedure is executed in four sequential phases.

Phase 1: Feature Discretization

To evaluate continuous and categorical variables uniformly, every continuous numeric feature is discretized into ordinal bins using equal-frequency (quantile) binning. This ensures that the downstream categorical association tests remain valid and resilient to outliers, while categorical features are left unchanged.

Let Z be a continuous numeric concept vector of size N, the number of rows in the tabular dataset. The domain of Z is partitioned into $q = \min(4, |\mathcal{U}(Z)|)$ disjoint intervals, where $\mathcal{U}(Z)$ denotes the set of unique values of Z. Using the empirical quantiles $Q_0, \dots, Q_q$, where $Q_0 := \min(Z)$ and $Q_q := \max(Z)$, the bins I_r are defined as

$$I_r = \begin{cases} [Q_{r-1}, Q_r) & \text{for } r \in \{1, \dots, q-1\}, \\ beginequation2pt][Q_{q-1}, Q_q] & \text{for } r = q. \end{cases} \tag{5.2}$$

Phase 2: Pairwise Association Testing via Cramér's V

Following [BPT+23], an $R_i \times R_j$ contingency table $\mathbf{O}$ is constructed for every pair of discretized concepts (C_i, C_j), where $R_i := |\mathcal{U}(C_i)|$ and $R_j := |\mathcal{U}(C_j)|$. The entry $O_{u,v}$ represents the observed joint frequency of category u of C_i and category v of C_j.

Under the null hypothesis H_0 of independence, the expected frequency $E_{u,v}$ for cell (u, v) is computed as

$$E_{u,v} = \frac{\left(\sum_{b=1}^{R_j} O_{u,b}\right)\left(\sum_{a=1}^{R_i} O_{a,v}\right)}{n}, \tag{5.3}$$

where n denotes the total sample size. The Pearson chi-squared (χ^2) statistic is then defined as

$$\chi^2 = \sum_{u=1}^{R_i}\sum_{v=1}^{R_j} \frac{(O_{u,v} - E_{u,v})^2}{E_{u,v}}. \tag{5.4}$$

The statistical significance $p_{i,j}$ is derived from the χ^2 cumulative distribution function with $\mathrm{dof} = (R_i - 1)(R_j - 1)$ degrees of freedom.

To quantify the strength of association independently of sample size and table dimensions, Cramér's V is computed, denoted A to avoid a notational conflict with the rest-of-question text V used elsewhere in this thesis:

$$A(C_i, C_j) = \sqrt{\frac{\chi^2}{n \cdot (\min(R_i, R_j) - 1)}}, \tag{5.5}$$

where $A(C_i, C_j) \in [0, 1]$, with 0 indicating independence and 1 indicating perfect association.

Phase 3: Association Filtering

To eliminate weak or spurious relationships, a pair (C_i, C_j) is retained if and only if it satisfies a dual-threshold condition:

1. **Statistical significance:** $p_{i,j} < \alpha \quad (\alpha = 0.005)$
2. **Effect size:** $A_{i,j} \geq A_{\min} \quad (A_{\min} = 0.3)$

Surviving pairs are sorted by descending association strength to form a prioritized candidate queue:

$$\mathcal{E} = \{(C_i, C_j, A_{i,j}) \mid p_{i,j} < \alpha \wedge A_{i,j} \geq A_{\min}\}\,. \tag{5.6}$$

Phase 4: Constrained Maximum Spanning Forest Partitioning

Let $\mathcal{C} = \{C_1, \ldots, C_M\}$ denote the static, dataset-wide set of concepts corresponding to the table columns. Unlike in natural-language datasets, $\mathcal{C}$ remains identical for every instance x.

The partitioning of $\mathcal{C}$ is modeled as finding a **Constrained Maximum Spanning Forest** on an undirected graph $G = (\mathcal{C}, \mathcal{E}, W)$, with edge weights $w(C_i, C_j) = A(C_i, C_j)$. This partitioning is resolved via a modified Kruskal's algorithm constrained by a maximum component size $n_{\max}$:

Algorithm 1 Constrained Greedy Partitioning

Initialize partition $\mathcal{P} \leftarrow \{\{C\} \mid C \in \mathcal{C}\}$ ▷ All concepts start as singletons
Sort $\mathcal{E}$ descending by weight $A_{i,j}$
for each edge $e = (C_i, C_j, A_{i,j}) \in \mathcal{E}$ **do**
  Find sets $P_a, P_b \in \mathcal{P}$ containing C_i and C_j
  if $P_a \neq P_b \wedge |P_a \cup P_b| \leq n_{\max}$ **then**
    $\mathcal{P} \leftarrow (\mathcal{P} \setminus \{P_a, P_b\}) \cup \{P_a \cup P_b\}$ ▷ Merge groups
  end if
end for
return $\mathcal{P}$

The resulting partition $\mathcal{P}$ serves as the tabular oracle output $S := \mathcal{P}$, which is computed once for the entire dataset. Downstream steps, namely joint counterfactual generation and Shapley-based redistribution (Section 5.2), then proceed identically to the natural-language pipeline, with each block $P \in \mathcal{P}$ used as a correlation group $s \in S$.

5.1.4 Putting Joint Interventions into Practice

With an oracle $\mathcal{O}(x, C, n_{\max})$ established that groups concepts into a set S based either on an LLM's judgment or on statistical evidence derived from the dataset, it becomes possible to intervene on the selected concepts jointly. Because the original metric of Matton et al. [MNG24] already contains the procedures required to intervene on a single concept, only minimal adaptation of the existing implementation is required. Most notably, the counterfactual generator must be updated to support interventions on multiple concepts simultaneously, enabling the measurement of the joint effect that these concepts have, as introduced in Section 4.2.

It is, however, not sufficient to simply intervene on all concepts within a correlation set $s \in S$ combined: doing so alone would make it impossible to distinguish the individual contributions, discussed further in Section 5.2, of each concept to the joint effect. Measuring only the effect of all concepts in a set acting together (the grand coalition) alongside the effect of each individual concept on its own is likewise insufficient. If a set contains more than two concepts, an individual concept may, for instance, behave differently on its own or within the grand coalition than when combined with only one other concept, as illustrated in Example **??**.

Example 5.2

Suppose an LLM answers whether a patient is likely suffering from bacterial meningitis, with $S = \{\{C_1, C_2, C_3\}\}$.

- C_1: High fever
- C_2: Neck stiffness
- C_3: Positive lumbar puncture (test result)

The model behaves roughly as follows:

- Removing high fever alone barely changes the diagnosis, because neck stiffness and the lumbar puncture result already provide sufficient evidence.
- Removing high fever together with neck stiffness, however, causes a much larger change than expected, because the remaining lumbar puncture result alone is less convincing and could, for instance, hint at multiple sclerosis instead.
- Removing all three concepts naturally changes the prediction completely.

Thus, the contribution of high fever depends strongly on whether neck stiffness is still present, even though this dependence is much weaker in the grand coalition. Consequently, the effect of intervening on C_1 differs substantially depending on whether it occurs within $\{C_1\}$ (alone), $\{C_1, C_2\}$, or the grand coalition $\{C_1, C_2, C_3\}$.

It is therefore necessary to investigate the shift in the model's predictions for all $2^{|s|} - 1$ possible non-empty combinations of concepts within a correlation set s. Using the slightly adapted procedures established by Matton et al. [MNG24], a counterfactual is created for each of these combinations. The evaluation pipeline then continues by collecting model responses $r = (y, e)$, consisting of the prediction y and the explanation e; the response-collection step itself requires no modification.

5.2 Adapting the Causal Effect: Introducing Shapley Values

5.2.1 Motivation

Having established how to identify correlated concepts and how to intervene on them jointly, it remains unresolved how to attribute the resulting causal effect to the observed shift in the prediction y.

Notably, the definition of causal effect by Matton et al. [MNG24] (see Equation 2.1) is defined only for individual concepts. Assigning a single CE value to a group of multiple concepts is undesirable for two reasons. First, doing so would make it nearly impossible to trace the effect of individual concepts, thereby forfeiting one of the metric's principal strengths and violating Desideratum **D1**. Second, it would require extensive reworking of the remainder of the evaluation pipeline: because the vectors of causal effects and explanation-implied effects must be of equal length[3] for the Pearson correlation to be computed, reducing the length of the CE vector would also require the EE vector to be reduced accordingly. This would introduce new sources of error and make it impossible to build directly on the foundation of the original metric. This option is therefore ruled out: the CE vector should still contain exactly $|C|$ entries. The remaining question is how the joint effects can be decomposed into individual contributions.

Naive approaches exist, such as computing the average effect of a concept across all coalitions in which it participates, ranging from the singleton case up to the grand coalition. This is problematic, however, since a concept's own effect is heavily influenced by the impact of the other concepts present in the coalition. Fortunately, devising a new solution from scratch is unnecessary, since Shapley values from cooperative game theory address exactly this scenario.

5.2.2 Shapley Values

Shapley values were introduced by Lloyd Shapley in 1953 [Sha53]. Originating in cooperative game theory, they determine how to fairly distribute a total payout among a set of players based on each player's individual contribution to the coalition. A player's allocation is calculated by evaluating its expected marginal contribution across all possible coalitions that the player could form with the remaining players [Sha53; MCFH22; PF24].

Given a value function $v(T)$ defined for any subset $T \subseteq N$, where N denotes the set of all players, the value allotted to a player $m \in N$ is given by Equation 5.7.

$$\phi_m(v) = \sum_{T \subseteq N \setminus \{m\}} \frac{|T|!\,(|N| - |T| - 1)!}{|N|!} \Big[v(T \cup \{m\}) - v(T) \Big] \tag{5.7}$$

where $\frac{|T|!\,(|N|-|T|-1)!}{|N|!}$ denotes the weighting factor, and $v(T \cup \{m\}) - v(T)$ denotes the marginal contribution provided by player m.

Shapley proved that this solution is unique in satisfying four core axioms [Sha53; MCFH22; PF24]:

- **Efficiency**: The sum of the individual values equals the total value created by the grand coalition ($\sum_{i \in N} \phi_i(v) = v(N)$).

[3]The length of both vectors corresponds to the number of concepts under investigation, i.e., typically $|C|$ for the concepts present in the question x.

- **Symmetry**: Identical players who contribute the exact same amount to any coalition receive identical payouts (if $v(T \cup \{i\}) = v(T \cup \{j\})$ for all $T \subseteq N \setminus \{i, j\}$, then $\phi_i(v) = \phi_j(v)$).
- **Dummy**: A player who adds zero value to every coalition receives zero payout (if $v(T \cup \{m\}) - v(T) = 0$ for all $T \subseteq N \setminus \{m\}$, then $\phi_m(v) = 0$).
- **Additivity**: If two independent games are played, a player's payout from the combined games equals the sum of their payouts from each individual game ($\phi_m(v+w) = \phi_m(v) + \phi_m(w)$).

It must be noted, however, that the computational cost of Shapley values also grows exponentially with the size of the coalition [PF24]. Since $n_{\max}$ was already introduced during the counterfactual generation step in Section 5.1, this is not a major concern in practice, as the primary computational bottleneck lies in collecting LLM responses for the exponentially growing number of counterfactuals, rather than in computing the Shapley values themselves. However, should larger coalitions become necessary, approximation methods such as Monte Carlo sampling are available for the Shapley values [MCFH22].

Shapley values have a long history of use within the field of *Explainable AI*, most notably in *SHAP*, where they are used to calculate the contribution of an input feature to the final prediction [LL17]. They have also been applied to analyze natural language explanations, as in the *CC-SHAP* framework [PF24], already introduced in Section 2.2.2.

5.2.3 Implementing Shapley Values into the *Causal Effect* Definition

The previous section left two questions unanswered: how the value function v should be defined, and how Shapley values can practically be integrated into the proposed framework.

It is natural to build on Equation 2.1 for estimating the causal effect, as doing so helps preserve backward compatibility (Desideratum **D2**). However, this equation has so far only been defined for a single intervened concept. Moreover, Section 5.2.1 already established the decision not to change the method signature $CE(x, C_m)$: the causal effect should remain defined for a single concept. The solution is therefore to split the equation into two parts: the right-hand side of Equation 2.1 is repurposed as the value function of the Shapley game, and the resulting CE value is obtained as the payout allotted to the concept by the Shapley function.

The value function must be adapted slightly to support joint interventions on multiple concepts. First, let T be a set of concepts, and let $\mathbb{C}_i$ denote the set of all possible values (the full domain) for a concept $C_i \in T$. The set of all reasonably possible joint values for T, denoted $\mathbb{T}$, is the Cartesian product of these individual value sets, $\mathbb{T} = \prod_{C_i \in T} \mathbb{C}_i$, with $t \in \mathbb{T}$ denoting the tuple of the original values of the concepts in T, i.e., $t = (c_i)_{C_i \in T}$. Analogous to the notation of Matton et al., $\mathbb{T}' := \mathbb{T} \setminus \{t\}$ denotes the set of possible counterfactual joint values, the multi-concept analogue of $\mathbb{C}'_m$. The value function of the Shapley game for a set T is then defined analogously to Equation 2.1 as

$$v(T) = \frac{1}{|\mathbb{T}'|} \sum_{t' \in \mathbb{T}'} D_{\mathrm{KL}} \left(\mathbb{P}_M(Y \mid x_{t \to t'}) \parallel \mathbb{P}_M(Y \mid x) \right). \tag{5.8}$$

With the value function now defined, the equation for estimating the causal effect can be formulated. As in the original metric, the causal effect for concept C_m and question x is given in Equation 5.9. Note that $C_m \in s$, where the assignment of C_m to the set s follows the procedure described in Section 5.1.

$$CE(x, C_m, s) = \sum_{T \subseteq s \setminus \{C_m\}} \frac{|T|!\,(|s| - |T| - 1)!}{|s|!} \Big[v(T \cup \{C_m\}) - v(T) \Big] \tag{5.9}$$

Recall, however, that Matton et al. [MNG24] estimate faithfulness via the Pearson correlation coefficient (Equation 2.3), in which the causal effect is invoked as $CE(x, C)$. A version without s as an explicit parameter must therefore be constructed, receiving instead the set of concepts C present in the question. The question-level CE vector no longer depends explicitly on C_m and s. As in Matton et al. [MNG24], let C denote the set of concepts present in the question x. A helper function is introduced in which $n_{\max}$ for $\mathcal{O}$ is fixed to a constant value:

$$f_C(C_m) = s \in \mathcal{O}(x, C, 5) \quad \text{such that} \quad C_m \in s \tag{5.10}$$

Querying this helper yields the set s to which concept C_m belongs:

$$CE(x, C) = \Big(CE\big(x, C_m, f_C(C_m)\big) \Big)_{m=1}^{|C|} \tag{5.11}$$

This construction therefore yields a causal-effect definition compatible with the Pearson correlation coefficient used in the original metric of Matton et al. [MNG24].

5.3 Normalizing the Shapley-CE via Correction

5.3.1 The Problem: Systematic Shrinkage of Joint Effects

Equations 5.9 and 5.10 technically complete the evaluation pipeline. Running an actual implementation, however, reveals a systematic pattern in the resulting scores. Whenever a concept C_m is assigned to a correlation group s with $|s| > 1$, its resulting Shapley-CE tends to be lower than the causal effect that would have been measured had C_m been evaluated on its own, i.e., on average $CE(x, C_m, s) < CE(x, C_m, \{C_m\})$, even for concepts that are otherwise independent in the sense of Section 3.4.

This is a direct consequence of the Shapley *Efficiency* axiom (Section 5.2): the sum of payouts to all concepts in s always equals $v(s)$, the joint effect of intervening on the entire group at once. If the joint effect of a group is systematically smaller than the sum of what its members would achieve individually, Efficiency guarantees that this shortfall is distributed across the individual payouts $CE(x, C_m, s)$. Thus, every concept in a correlated group is, on average, "punished" for being grouped, regardless of whether it actually interacts with the other members of s.

Intuitively, the underlying mechanism resembles a ceiling effect. Suppose that intervening on Concept A alone shifts the model's predicted probability of outcome b from 50% to 80%, and that intervening on Concept B alone produces the same shift. Jointly intervening on both concepts cannot push the prediction to 160%, since probabilities are bounded by 100%; some shortfall relative to the naive sum is therefore unavoidable, independently of whether the two concepts interact.

Specifically, this ceiling is enforced during the Bayesian hierarchical modeling step presented by Matton et al. [MNG24] in Appendix C.2 of their work. To reconstruct the probability distribution vector $p_{y|I_{C^{\mathbf{x}}_{m'}}}$ from model responses Y, they employ a softmax function:

$$p_{y|I_{C^{\mathbf{x}}_{m'}}} = \frac{e^{\theta_{y|I_{C^{\mathbf{x}}_{m'}}}}}{\sum_{y \in \mathcal{Y}} e^{\theta_{y|I_{C^{\mathbf{x}}_{m'}}}}} \quad y \in \mathcal{Y} \tag{5.12}$$

Without detailing the surrounding variables exhaustively, its key property is that outputs are strictly confined to $[0, 1]$ and sum to exactly 1 across all possible predictions in $\mathcal{Y}$. This preserves the fundamental axiom that no set of mutually exclusive events can have a combined probability exceeding 100%.

Because of this strict normalization, the probability distribution resulting from a joint intervention cannot equal the sum of the distributions from individual interventions. If one were to naively add the isolated resulting distributions of an intervention on Concept A and an intervention on Concept B, the sum over the prediction space would equal 2. Since any valid probability distribution must sum to exactly 1, the joint intervention is inherently non-additive in probability space.

Beyond this extreme case of distributions summing to 2, this non-additivity also applies locally to individual predictions as a "soft ceiling" that continuously dampens joint effects. Because the softmax function applies an exponential transformation divided by a sum of exponentials, additive increases in the underlying model scores (logits) translate into sub-additive increases in probability. Example **??** illustrates this mechanism in detail, demonstrating how two independent concepts that individually shift a prediction by +20% and +30% yield a joint shift of only +44%, rather than the naive additive sum of +50%.

Example 5.3

Scenario: Let the prediction space consist of two outcomes, $\mathcal{Y} = \{y_1, y_2\}$. Suppose the model's baseline internal scores (logits) before any intervention are $\theta_{y_1} = \ln(4)$ and $\theta_{y_2} = \ln(6)$.
Using the softmax formula (Equation 5.12), the baseline probability for y_1 is:

$$P(y_1) = \frac{e^{\ln(4)}}{e^{\ln(4)} + e^{\ln(6)}} = \frac{4}{4+6} = 0.40 \quad (40\%) \tag{5.13}$$

Individual Interventions: Assume that intervening on a concept acts additively in the model's logit space.

- **Concept A:** Increases the logit of y_1 by $\Delta_A = \ln(2.25)$.

$$P^{(A)}(y_1) = \frac{4 \cdot e^{\ln(2.25)}}{(4 \cdot e^{\ln(2.25)}) + 6} = \frac{4 \cdot 2.25}{9+6} = \frac{9}{15} = 0.60 \quad (60\%) \tag{5.14}$$

Individual effect of A: $60\% - 40\% = +\mathbf{20}\%$

- **Concept B:** Independently increases the logit of y_1 by $\Delta_B = \ln(3.5)$.

$$P^{(B)}(y_1) = \frac{4 \cdot e^{\ln(3.5)}}{(4 \cdot e^{\ln(3.5)}) + 6} = \frac{4 \cdot 3.5}{14+6} = \frac{14}{20} = 0.70 \quad (70\%) \tag{5.15}$$

Individual effect of B: $70\% - 40\% = +\mathbf{30}\%$

Naive Expected Joint Effect: If probabilities were perfectly additive, intervening on A and B together would shift the probability by $20\% + 30\% = +\mathbf{50}\%$, yielding a final probability of 90%.

Actual Joint Intervention (The Soft Ceiling): When intervening on both concepts, their logit changes combine additively ($\Delta_A + \Delta_B = \ln(2.25) + \ln(3.5) = \ln(7.875)$). The softmax function computes the new joint probability as:

$$P^{(A,B)}(y_1) = \frac{4 \cdot e^{\ln(7.875)}}{(4 \cdot e^{\ln(7.875)}) + 6} = \frac{4 \cdot 7.875}{31.5 + 6} = \frac{31.5}{37.5} = 0.84 \quad (84\%) \tag{5.16}$$

Actual joint effect: $84\% - 40\% = +\mathbf{44}\%$

Conclusion: Due to the growing denominator in the softmax function, the actual joint effect (+44%) falls short of the naive additive sum (+50%).

Because the joint effect serves as the basis for calculating individual payouts, this inherent non-additivity guarantees that independent concepts are penalized for being placed in a coalition. Mathematically, this manifests as $v(s) < \sum_{C_m \in s} v(\{C_m\})$. This structural penalty provides a poor basis for further evaluation, necessitating an adjustment. However, removing the softmax function or altering the Bayesian sampling process would disrupt backward compatibility (Desideratum **D2**). The preferred solution is therefore to apply a normalization that counteracts this shrinkage, calibrated against the magnitude of the same effect observed for comparable treatments.

5.3.2 The Payout Coefficient

For a treatment $T \subseteq s$ with $T \neq \emptyset$, the *payout coefficient* is defined as

$$\gamma(x, T) := \frac{v(T)}{\sum\limits_{C_m \in T} v(\{C_m\})}, \tag{5.17}$$

the ratio of the group's joint effect to the sum of its members' singleton effects.[4] By definition, $\gamma(x, T) = 1$ indicates perfect additivity, $\gamma(x, T) < 1$ the shrinkage documented above, and $\gamma(x, T) > 1$ reinforcement. For a singleton treatment $T = \{C_m\}$, numerator and denominator coincide, so $\gamma(x, \{C_m\}) \equiv 1$: the correction defined below therefore leaves singleton effects, and with them Desideratum **D2**.1, untouched.

5.3.3 Constructing a Reference Set

A treatment's own coefficient $\gamma(x, T)$ cannot be used to correct its own value function: doing so would force $v(T)/\gamma(x, T) = \sum_{C_m \in T} v(\{C_m\})$ identically for every treatment, discarding exactly the information about genuine concept interaction that Desideratum **D4** requires the metric to retain. Instead, $\gamma(x, T)$ is corrected using the *typical* coefficient observed for treatments of the same size $|T|$ elsewhere in the dataset.

[4] If $\sum_{C_m \in T} v(\{C_m\}) = 0$, every concept in T has zero singleton effect; in this degenerate case $\gamma(x, T) := 1$ is set by convention, so that no correction is applied.

Let $S := \mathcal{O}(x, C, n_{\max})$ denote the correlation groups for question x. Three disjoint pools of reference treatments matching $|T|$ in size are considered, in descending priority:

$$\mathcal{R}_1(x, s, T) := \Big\{ (x, T'') \ : \ T'' \subseteq s, \ T'' \neq T, \ |T''| = |T| \Big\}, \tag{5.18}$$

$$\mathcal{R}_2(x, s, T) := \Big\{ (x, T'') \ : \ T'' \subseteq s', \ s' \in S \setminus \{s\}, \ |T''| = |T| \Big\}, \tag{5.19}$$

$$\mathcal{R}_3(x, s, T) := \Big\{ (x'', T'') \ : \ x'' \in X \setminus \{x\}, \ T'' \subseteq s'', \ s'' \in \mathcal{O}(x'', C, n_{\max}), \ |T''| = |T| \Big\}. \tag{5.20}$$

$\mathcal{R}_1$ draws treatments of matching size from within the same correlation group s; $\mathcal{R}_2$ widens the search to other groups within the same question x; $\mathcal{R}_3$ widens it further to the entire dataset X. By construction, the three pools are pairwise disjoint. The reference set is obtained by drawing $n_{\text{ref}} := 5$ elements greedily from $\mathcal{R}_1 \cup \mathcal{R}_2 \cup \mathcal{R}_3$, exhausting each pool before moving to the next:

$$\mathcal{R}(x, s, T) \subseteq \mathcal{R}_1 \cup \mathcal{R}_2 \cup \mathcal{R}_3, \qquad |\mathcal{R}(x, s, T)| = \min\Big(n_{\text{ref}}, \ |\mathcal{R}_1 \cup \mathcal{R}_2 \cup \mathcal{R}_3|\Big), \tag{5.21}$$

with ties within a pool broken arbitrarily, e.g., uniformly at random. If $\mathcal{R}_1 \cup \mathcal{R}_2 \cup \mathcal{R}_3 = \emptyset$, no reference treatment of matching size exists anywhere in the dataset, and no correction is applied (Section 5.3.4).

5.3.4 The Corrected Value Function

The mean payout coefficient over the reference set,

$$\bar{\gamma}(x, s, T) := \begin{cases} \dfrac{1}{|\mathcal{R}(x, s, T)|} \displaystyle\sum_{(x'', T'') \in \mathcal{R}(x, s, T)} \gamma(x'', T'') & \text{if } \mathcal{R}(x, s, T) \neq \emptyset, \\ \mathit{beginequation3mm]}1 & \text{if } \mathcal{R}(x, s, T) = \emptyset, \end{cases} \tag{5.22}$$

estimates the typical shrinkage or amplification expected for treatments of size $|T|$, independently of T's own realization. The corrected value function is

$$v_{\text{adj}}(T) := \frac{v(T)}{\bar{\gamma}(x, s, T)}, \qquad T \neq \emptyset, \tag{5.23}$$

with $v_{\text{adj}}(\emptyset) := v(\emptyset) = 0$ by convention.[5]

Substituting v_{adj} for v in Equation 5.9 yields the corrected causal effect

$$CE_{\text{adj}}(x, C_m, s) := \sum_{T \subseteq s \setminus \{C_m\}} \frac{|T|!\,(|s| - |T| - 1)!}{|s|!} \Big[v_{\text{adj}}(T \cup \{C_m\}) - v_{\text{adj}}(T)\Big], \tag{5.24}$$

which replaces $CE(x, C_m, s)$ from this point onward in populating the causal-effect vector $CE(x, C)$ of Equation 5.10.

In summary, the correction is designed to remove, in expectation, exactly the structural non-additivity established in Section 5.3.1, while leaving genuine interaction effects (Section 3.4) intact for any individual treatment. Furthermore, it preserves backward compatibility (Desideratum **D2**) for dataset averages, meaning that average CE scores match those obtained under the original metric.

[5]Recall from Equation 5.8 that $\mathbb{T}' = \emptyset$ when $T = \emptyset$, so $v(\emptyset)$ is not defined by the formula itself; $v(\emptyset) := 0$ is the natural convention, already used implicitly by the $T = \emptyset$ term of the Shapley sum in Equation 5.9, and is stated explicitly here for completeness.

5.4 Remaining Evaluation Pipeline

The previous sections introduced the modified formulation of the *Causal Effect* (Equation 5.24), which is now computed using Shapley values. Furthermore, the underlying value function was extended to compensate for the systematic underestimation of joint concept effects discussed in Section 5.3.

After the CE vector has been computed, two major stages remain in the evaluation pipeline proposed by Matton et al. [MNG24]: estimating the *Explanation Implied Effect* (EE) and computing the final faithfulness score from the relationship between both vectors. Fortunately, neither of these components requires any modification.

The computation of the EE vector remains unchanged because the output of the modified causal-effect estimation is still defined on a per-concept basis. Consequently, the adjusted CE vector is fully compatible with the inputs and outputs expected by the original EE estimation procedure. Although *FELICS* explicitly models interactions between concepts, the objective of the EE vector remains identical: estimating which concepts the evaluated LLM considers influential enough to mention in its explanation.

For example, two concepts may each have only a minor individual influence on the prediction while jointly producing a substantial change due to a synergistic interaction. In this case, neither concept would be expected to appear frequently in explanations when considered individually, whereas both should be referenced once their combined effect becomes significant. Likewise, in a redundant setting, both concepts should still be mentioned whenever their joint contribution remains important for the prediction. Conversely, if one concept contributes only marginally to the coalition's overall effect, it should also appear less frequently in the explanation. Consequently, the alignment between causal effects and explanation-implied effects remains an appropriate indicator of explanation faithfulness.

Since the EE estimation remains unchanged, the final faithfulness estimation can likewise be adopted without modification. This applies both to the formal definition based on the *Pearson Correlation Coefficient* (Equation 2.3) and to the practical estimation using *Bayesian Hierarchical Modelling*, as proposed by Matton et al. [MNG24]. Furthermore, the computation of *Dataset-Level Faithfulness* (Equation 2.4) is unaffected. By restricting all modifications to the estimation of causal effects, *FELICS* builds upon the validated components of the original framework while minimizing the number of changes required throughout the remaining evaluation pipeline.

5.5 Analysis

5.5.1 Reviewing Changes

Having introduced the individual modifications of *FELICS*, it is useful to summarize how the proposed framework differs from the original *Causal Concept Faithfulness Metric* by Matton et al. [MNG24]. Rather than repeating the original evaluation pipeline presented in Section 2.3 and each modification introduced throughout this chapter, Figure 5.1 provides a side-by-side comparison of both approaches.

The diagram is organized into three columns. The left column contains the original evaluation pipeline proposed by Matton et al. [MNG24], whereas the right column illustrates the corresponding components introduced by *FELICS*. Between them, the shared components are shown to emphasize which parts of the framework remain unchanged. Whenever a processing step differs depending on the underlying dataset type (natural-language or tabular), the corresponding alternatives are separated by the • symbol. Finally, the arrows indicate how intermediate variables and results propagate through the evaluation pipeline and serve as inputs to subsequent processing stages.

5.5.2 Theoretical Review of Desiderata

Section 4.1 introduced five desiderata that the proposed extension should satisfy. These range from preserving the strengths and compatibility of the original *Causal Concept Faithfulness Metric* (Desiderata **D1** and **D2**) to explicitly addressing the *Correlated Concepts Problem* (Desiderata **D3** and **D4**) while maintaining practical computational feasibility (Desideratum **D5**).

At this stage, the conceptual design of *FELICS* is complete, whereas its empirical evaluation on a reference implementation are presented in the following chapter. It is therefore appropriate to revisit these desiderata and assess whether they are satisfied by the theoretical construction of the proposed framework.

D1 Preserving the strengths of the *Causal Concept Faithfulness Metric*

The principal strengths of the metric proposed by Matton et al. [MNG24] stem from its operation on high-level semantic concepts rather than individual tokens and from its ability to estimate causal and explanation-implied effects for every concept separately. Since *FELICS* retains this concept-based representation and continues to compute both the CE and EE vectors on a per-concept basis, these strengths are preserved without modification.

D2 Backward compatibility

A second desideratum requires the proposed extension to remain compatible with the original metric. More specifically, Desideratum **D2**.1 states that whenever every correlation group contains exactly one concept, the resulting causal-effect estimates should be identical to those obtained by Matton et al. [MNG24]. This follows directly from the properties of the normalization factor and the Shapley-value formulation, as demonstrated in Math Box **??**.

Desideratum **D2**.2 considers the more general case in which concepts are grouped together but remain mutually independent. Strictly speaking, the formulation stated in Section 4.1 cannot be satisfied exactly, since the probability normalization imposed by the softmax function introduces the ceiling effect discussed in Section 5.3.1. Consequently, even perfectly independent concepts do not generally satisfy $v(s) = \sum_{C_m \in s} v(\{C_m\})$.

The normalization proposed in Section 5.3 therefore compensates for this systematic shrinkage only in a relative sense. A correlation group behaves independently whenever its payout coefficient equals the expected payout coefficient of equally sized groups within the dataset. In other words, the items in the group are independent if around half of equal-sized references interact more synergisticly and the other half more redundantly. Under this practically attainable notion of independence, no additional interaction effect is attributed to the group, and the resulting causal-effect estimates coincide with those of the original metric, as demonstrated in Math Box **??**.

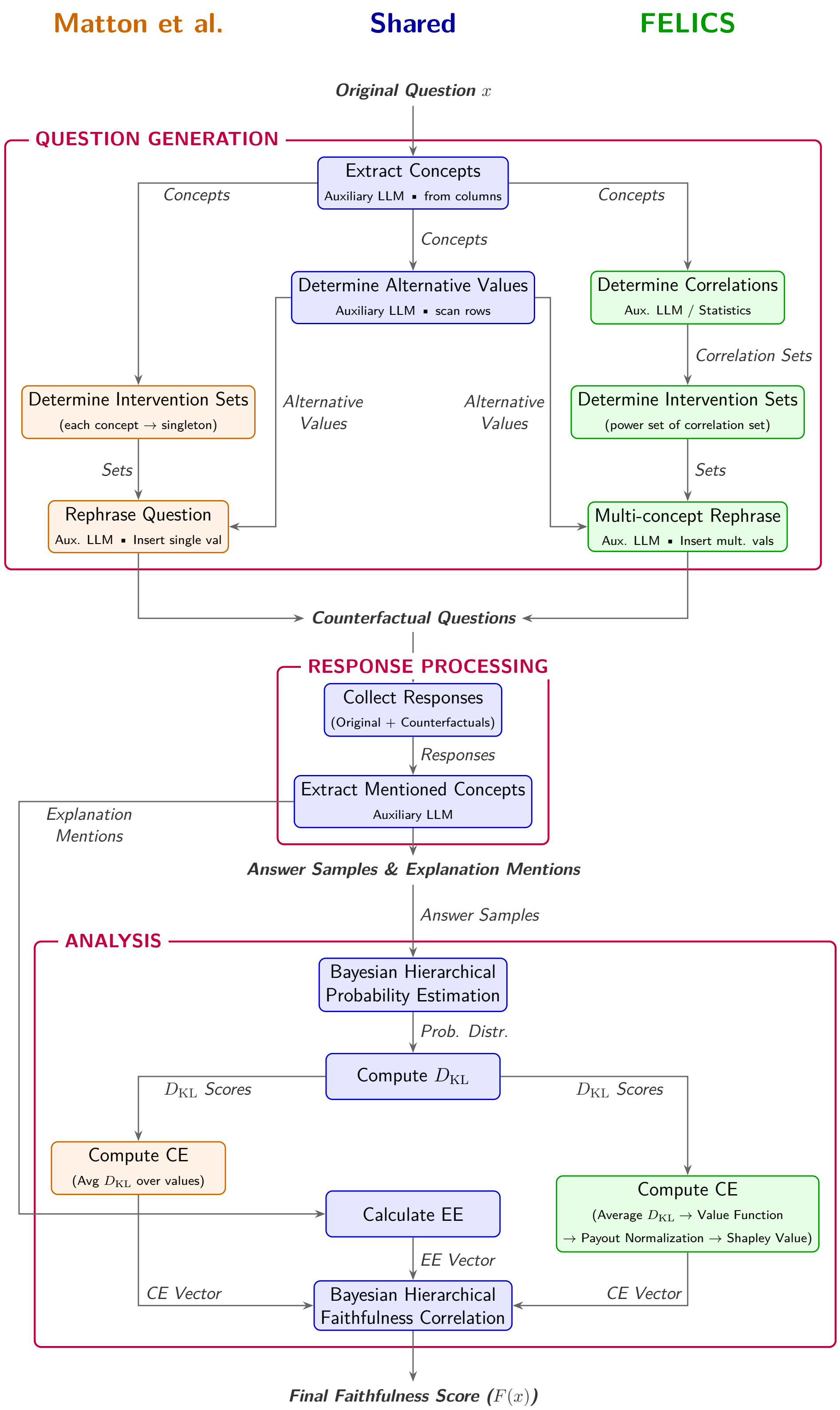


Figure 5.1: Comparison of the evaluation pipeline between Matton et al. [MNG24] (left) and the proposed *FELICS* framework (right), highlighting their shared foundations as well as the modifications to intervention generation and causal-effect estimation.

Mathematical Demonstration 5.1

Let the correlation group consist of exactly one concept,

$$s = \{C_m\}. \tag{5.25}$$

The payout coefficient introduced in Section 5.3 becomes

$$\gamma(x, s) = \frac{v(s)}{\sum_{C_i \in s} v(\{C_i\})} = \frac{v(\{C_m\})}{v(\{C_m\})} = 1. \tag{5.26}$$

Since the reference set used to compute the correction factor $\bar{\gamma}(x, s, T)$ is restricted to sets of identical treatment size, the only admissible reference size is likewise $|T| = 1$. Consequently,

$$\bar{\gamma}(x, s, T) = 1, \tag{5.27}$$

and the normalized value function reduces to

$$v_{\text{adj}}(T) = \frac{v(T)}{\bar{\gamma}(x, s, T)} = v(T). \tag{5.28}$$

The adjusted causal effect is then obtained from the Shapley formulation

$$CE_{\text{adj}}(x, C_m, s) = \sum_{T \subseteq s \setminus \{C_m\}} \frac{|T|!(|s| - |T| - 1)!}{|s|!} \left[v_{\text{adj}}(T \cup \{C_m\}) - v_{\text{adj}}(T)\right]. \tag{5.29}$$

Since $s = \{C_m\}$, the only admissible coalition is the empty set, $T = \emptyset$. The weighting coefficient therefore simplifies to

$$\frac{0!(1 - 0 - 1)!}{1!} = 1, \tag{5.30}$$

and using the convention $v_{\text{adj}}(\emptyset) = 0$,

$$CE_{\text{adj}}(x, C_m, s) = v_{\text{adj}}(\{C_m\}). \tag{5.31}$$

Because no normalization is applied for singleton groups,

$$v_{\text{adj}}(\{C_m\}) = v(\{C_m\}), \tag{5.32}$$

which is exactly the value function underlying the causal-effect definition of Matton et al. [MNG24]. Therefore,

$$CE_{\text{adj}}(x, C_m, \{C_m\}) = CE_{\text{Matton}}(x, C_m) \tag{5.33}$$

showing that *FELICS* reduces exactly to the original CE every correlation group contains only a single concept. Because the definitions of the explanation implied effect and faithfulness (see Equations 2.2 and 2.3) are unchanged, both metrics are equivalent in this scenario.

Mathematical Demonstration 5.2

Assume a correlation group s whose concepts are independent relative to the dataset. By definition of the normalization introduced in Section 5.3, this corresponds to

$$\bar{\gamma}(x, s, s) = \gamma(x, s). \tag{5.34}$$

The adjusted value of the grand coalition therefore becomes

$$v_{\text{adj}}(s) = \frac{v(s)}{\gamma(x, s)}. \tag{5.35}$$

Substituting the definition of the payout coefficient,

$$\gamma(x, s) = \frac{v(s)}{\sum_{C_i \in s} v(\{C_i\})}, \tag{5.36}$$

yields

$$v_{\text{adj}}(s) = \sum_{C_i \in s} v(\{C_i\}). \tag{5.37}$$

Hence, after normalization, the value assigned to the grand coalition equals the sum of the individual atomic intervention effects.
Assume furthermore that every concept contributes independently to every coalition. Then the marginal contribution of a concept is identical in every coalition in which it appears,

$$v_{\text{adj}}(T \cup \{C_m\}) - v_{\text{adj}}(T) = v(\{C_m\}), \qquad \forall T \subseteq s \setminus \{C_m\}. \tag{5.38}$$

Since the Shapley weights form a partition of unity,

$$\sum_{T \subseteq s \setminus \{C_m\}} \frac{|T|!(|s| - |T| - 1)!}{|s|!} = 1, \tag{5.39}$$

the adjusted causal effect reduces to

$$CE_{\text{adj}}(x, C_m, s) = v(\{C_m\}). \tag{5.40}$$

Finally, $v(\{C_m\})$ is precisely the KL-divergence-based intervention value used by Matton et al. [MNG24] to define the causal effect of an individual concept (see Equation 2.1). Consequently,

$$CE_{\text{adj}}(x, C_m, s) = CE_{\text{Matton}}(x, C_m), \tag{5.41}$$

showing that no additional causal effect is attributed to correlation groups whose concepts behave independently relative to the dataset.

D3 Avoiding the assumption of surgical interventions

The original metric proposed by Matton et al. [MNG24] fundamentally relies on the assumption that *surgical interventions* are possible. As discussed in Section 3.3, this assumption does not hold in many practical scenarios where concepts are inherently correlated. Consequently, removing this assumption constituted one of the primary design goals of *FELICS*.

Instead of intervening on individual concepts exclusively, *FELICS* performs joint interventions whenever concepts belong to the same correlation group. This principle was introduced in Section 4.2. Assuming that the oracle correctly identifies concepts that cannot be meaningfully intervened upon in isolation, the framework no longer depends on the existence of surgical interventions. Therefore, Desideratum **D3** can be regarded as fulfilled.

D4 Correctly capturing interaction effects

As established in Section **??**, concepts may exhibit independent, redundant, or synergistic interactions. The proposed framework should therefore correctly distinguish between these different forms of behavior and attribute their causal effects accordingly.

This property follows directly from the demonstration in Math Box **??**. If a correlation group behaves independently relative to the dataset, its payout coefficient equals the expected payout coefficient for groups of identical size,

$$\bar{\gamma}(x, s, s) = \gamma(x, s), \tag{5.42}$$

causing the adjusted causal effects to reduce to those of the original metric. Conversely, if a correlation group exhibits stronger redundancy or synergy than comparable groups, its payout coefficient deviates from this expected value,

$$\bar{\gamma}(x, s, s) \neq \gamma(x, s). \tag{5.43}$$

As a consequence, the normalization either decreases or increases the adjusted coalition value before the Shapley-value attribution is performed. Redundant interactions therefore satisfy

$$CE_{\text{adj}}(x, C_m, s) < CE_{\text{Matton}}(x, C_m), \tag{5.44}$$

whereas synergistic interactions satisfy

$$CE_{\text{adj}}(x, C_m, s) > CE_{\text{Matton}}(x, C_m). \tag{5.45}$$

The proposed framework thus behaves consistently with the intended interpretation of interaction effects and can therefore be argued to satisfy Desideratum **D4**.

D5 Maintaining computational feasibility

The proposed extension would provide little practical benefit if its computational cost increased to an impractical level.

Introducing joint interventions naturally increases the number of counterfactual questions that must be generated. In the worst case, evaluating every possible concept combination would require an exponential number of LLM calls. To limit this growth, the oracle introduced in Section 5.1 groups together only those concepts for which joint interventions are considered necessary. Furthermore, the mandatory upper bound $n_{\max}$ ensures that the number of generated interventions never exceeds the limit established in Equation 5.1.

Whether Desideratum **D5** is satisfied ultimately depends on what is considered computationally feasible in practice. Nevertheless, it can reasonably be argued that the proposed framework fulfills this requirement. Although *FELICS* is necessarily more computationally expensive than the original metric of Matton et al. [MNG24], this additional cost is restricted to questions containing correlated concepts. The combination of oracle-guided correlation detection and the hard upper bound on intervention size provides a principled trade-off between computational cost and modeling accuracy.

5.5.3 Expected Influence on Results

Since the original framework by Matton et al. [MNG24] cannot reliably handle concepts acting in a redundant or synergistic manner, their true causal effect is not captured. The influence exerted by *FELICS* on CE results relative to the *Causal Concept Faithfulness* metric is determined by the intervention mode, as discussed in Section 3.4.

Effects on CE Scores

The relationship between CE_{adj} and CE_{original} is governed by the payout coefficient γ relative to the reference-set average $\bar{\gamma}$. For a pair of concepts C_i, C_j in a group $s = \{C_i, C_j\}$, where

$$\gamma(C_i, C_j) = \frac{v(\{C_i, C_j\})}{v(\{C_i\}) + v(\{C_j\})},$$

the following cases are distinguished:

- **Redundancy ($\gamma(C_i, C_j) < \bar{\gamma}$):** The joint effect is smaller than the sum of singleton effects. When concepts are *added* to a question, $CE_{\text{adj}} < CE_{\text{original}}$ for both concepts (Matton overestimates). When concepts are *removed*, $CE_{\text{adj}} > CE_{\text{original}}$ (Matton underestimates).
- **Synergy ($\gamma(C_i, C_j) > \bar{\gamma}$):** The joint effect exceeds the sum of singleton effects. When concepts are *added*, $CE_{\text{adj}} > CE_{\text{original}}$ (Matton underestimates). When concepts are *removed*, $CE_{\text{adj}} < CE_{\text{original}}$ (Matton overestimates).
- **Independence ($\gamma(C_i, C_j) = \bar{\gamma}$):** The joint effect equals the sum of singleton effects. In this case, $CE_{\text{adj}} = CE_{\text{original}}$ for both concepts, and FELICS matches Matton's original metric.

Value-Replacement Interventions

When concept values are *modified* rather than added or removed, the relationship between CE_{adj} and CE_{original} depends on the direction of the effect. If concepts C_i and C_j exhibit minor influence before intervention but major individual influence afterward while also interacting redundantly post-intervention, Matton's metric overestimates their effect ($CE_{\text{adj}} < CE_{\text{original}}$). Conversely, if their influence is major before intervention but minor after being changed on their own, again with redundant behavior after intervention, Matton's metric underestimates ($CE_{\text{adj}} > CE_{\text{original}}$). This sign inversion mirrors the behavior observed for addition vs. removal interventions.

However, value-based interventions introduce additional complexity: modifying a concept's value may create contradictory information in the question (e.g., changing nationality from `Japanese` to `Indian` while retaining `Japanese` as the mother tongue). Such inconsistencies can obscure the measurement of causal effects, making it hard to predict the generalized impact of multi-concept modifications.

Despite these intervention-specific variations, the dataset-level average of all CE values is expected to remain approximately equal between both metric. This approximation holds because the reference-set construction ensures that each group has a comparable probability of being included in reference sets for other groups, and the Shapley *efficiency property* guarantees that no causal influence is systematically lost or gained. A formal demonstration of this approximation is provided in Appendix A.2.

Effects on Faithfulness Scores

As established in Section 5.4, the definition of Explanation-Implied Effect (EE) remains entirely unchanged between FELICS and the *Causal Concept Faithfulness* metric. Consequently, the EE vectors are identical for both metrics.

The faithfulness score, however, is still determined by the Pearson correlation coefficient between the CE and EE vectors. Since Pearson correlation depends on the magnitudes of the data points (unlike rank-based metrics such as Spearman), and the individual CE values may differ between FELICS and Matton's metric, the resulting faithfulness scores are not guaranteed to be equal. In fact, whenever interactions are present, the alignment between CE and EE vectors generally changes, except in the rare case where widening gaps for some concepts are exactly offset by narrowing gaps for others. This behavior is illustrated in an example in appendix A.1.

Still, two special cases are guaranteed to yield identical faithfulness scores: When every correlation group contains exactly one concept (Desideratum **D2**.1) and when all concepts are entirely independent (Desideratum **D2**.2).

Chapter 6
Evaluation Methodology

The preceding chapters have modeled the proposed extension *FELICS* from a purely theoretical standpoint: not only were the desiderata guiding its design established, but its behavior was also formalized and, in several cases, mathematically demonstrated to satisfy these desiderata. However, it remains to be shown whether the metric behaves in practice as its theoretical construction suggests. To this end, a thorough evaluation is conducted against a reference implementation, in which several central research questions are addressed and the resulting data and findings are processed and visualized.

6.1 Research Questions

To empirically evaluate the efficacy, theoretical soundness, and practical implications of the proposed *FELICS* framework, the experimental design pursued in this chapter is guided by five research questions, each formulated as a testable hypothesis. Taken together, these hypotheses assess whether the implementation successfully fulfills the formalized desiderata (see Section 4.1) and gauge which behaviors the metric extension exhibits in practice.

RQ1 Impact of Correlation Strength on Causal Effect Estimates: Does explicitly modeling concept interactions introduce a significant difference in Causal Effect (CE) estimates whenever the concepts within a question are more strongly correlated?
Hypothesis: As established in Section 5.1, the oracle $\mathcal{O}(x, C, n_{\max})$ partitions the concept set C of a question x into correlation groups, with larger or more numerous groups indicating that the oracle has judged the underlying concepts to be more strongly correlated. Increasing $n_{\max}$ permits larger groups and therefore allows more of this assumed correlation to be captured by FELICS. It is hypothesized that the resulting difference in CE estimates between the atomic configuration ($n_{\max} = 1$) and a joint configuration ($n_{\max} > 1$) correlates positively with the corresponding difference in correlation strength. Since FELICS reduces exactly to the original *Causal Concept Faithfulness* metric whenever $n_{\max} = 1$ (Desideratum **D2**.1), this comparison can be conducted entirely within FELICS by contrasting its atomic and joint configurations, without requiring a separate evaluation run of Matton et al.'s [MNG24] original implementation.

RQ2 Convergence to the Baseline Metric: Does FELICS, when restricted to atomic interventions, converge to the original *Causal Concept Faithfulness* metric not only in theory but also in practically observed results?
Hypothesis: Math Box **??** shows that whenever every correlation group contains exactly one concept, the causal-effect formula of FELICS reduces exactly to that of Matton et al.'s [MNG24] original metric (Desideratum **D2**.1). Because FELICS is nevertheless evaluated using a separate, independently executed implementation rather than by direct algebraic substitution into Matton et al.'s reported results, this theoretical equivalence must still be confirmed empirically: discrepancies could in principle arise from accidental changes that break compatibility. It is therefore hypothesized that FELICS with $n_{\max} = 1$ and the original *Causal Concept Faithfulness* metric do not produce statistically distinguishable

results, evaluated first at the level of per-concept CE estimates and second at the level of the resulting question-level faithfulness scores.

RQ3 Preservation of Within-Question Concept Rankings: Does the payout-coefficient correction applied by FELICS preserve the relative ranking of concepts within a question, or does it reshuffle which concept appears most responsible for an unfaithful explanation?
Hypothesis: Desideratum **D1** requires FELICS to preserve Matton et al.'s [MNG24] ability to reveal interpretable, question-level patterns of unfaithfulness. This requirement extends beyond matching aggregate CE or faithfulness values: the corrected metric must continue to tell a coherent story about which concepts drive unfaithfulness within an individual question. It is hypothesized that, within a question, the rank order of concepts obtained from the adjusted causal effect CE_{adj} correlates strongly and positively with the rank order obtained from the raw, uncorrected Shapley causal effect CE_{raw}—that is, the payout-coefficient correction rescales the magnitude of each concept's estimated effect without substantially reordering which concepts appear most responsible. Should this hypothesis be rejected, the resulting loss of per-question interpretability would remain a relevant cost to report, even if the aggregate equivalence targeted by **RQ2** and **RQ4** holds.

RQ4 Enhancement of Faithfulness Scores through Interaction Modeling: Does capturing interaction effects between concepts yield higher measured faithfulness for the evaluated models?
Hypothesis: Matton et al. [MNG24] report that most of the LLMs evaluated on their benchmark datasets already behave in a relatively faithful manner. However, because their original metric assumes disentangled, non-interacting concepts, it cannot capture cases in which a genuinely faithful model's explanation reflects the joint, rather than individual, influence of correlated concepts (cf. Section 5.4). This mismatch artificially widens the gap between the *Causal Effect* and the *Explanation-Implied Effect*, thereby penalizing an otherwise faithful model with an unfairly low faithfulness score. Since FELICS is designed to close exactly this gap, and given that evaluated LLMs can generally be assumed to be relatively faithful, it is hypothesized that question-level faithfulness scores are significantly higher once interaction effects are considered ($n_{\text{max}} > 1$) than under the atomic baseline ($n_{\text{max}} = 1$). As with **RQ1**, this comparison is conducted entirely within FELICS, relying on the equivalence between FELICS ($n_{\text{max}} = 1$) and Matton et al.'s [MNG24] original metric targeted by **RQ2**.

RQ5 Resolution of the Motivating Underestimation Example: Does FELICS resolve the specific underestimation identified as motivation for this thesis?
Hypothesis: As discussed in Section 3.3, Matton et al. [MNG24] identify a MedQA question in which the concepts *the patient's eating disorder* and *the patient's self-perception of weight* each receive a surprisingly low Causal Concept Effect, because the evaluated LLM can reconstruct either concept from the other whenever only one of them is removed. Section 3.4 establishes that, under the removal-based intervention scheme adopted for this dataset, this behavior constitutes redundancy, characterized by $CE(x, \{A, B\}) > CE(x, A) + CE(x, B)$ for the two concepts A and B. Consequently, it is hypothesized that FELICS's grouped, corrected estimate exceeds Matton et al.'s original estimate for both concepts, i.e., $CE_{\text{adj}}(A) > CE_{\text{original}}(A)$ and $CE_{\text{adj}}(B) > CE_{\text{original}}(B)$, and that this relationship holds consistently across all evaluated models.

6.2 Evaluation Settings

6.2.1 Reference Implementation

Thus far, *FELICS* has only been modeled mathematically, without an accompanying executable implementation. However, an implementation is required to automatically gather empirical data and to allow fellow researchers to use the FELICS pipeline without having to reimplement it themselves. A *reference implementation* was therefore developed, building largely upon the same code base as the original implementation by Matton et al. [MNG24] and introducing changes only where appropriate and necessary. It is available at `https://github.com/JonathanZopf/FELICS`. For comparison, the evaluation is additionally run against the original implementation, available at `https://github.com/kmatton/walk-the-tall`

6.2.2 Selection of Datasets

The evaluation is conducted on three distinct datasets. BBQ and MedQA not only represent natural language datasets with their own particular characteristics, but also enable a direct comparison between FELICS and Matton et al.'s [MNG24] original metric. New German Credit, by contrast, is a novel dataset that can only be evaluated with FELICS, since the original metric does not support tabular data; it represents a sample tabular dataset, offering the advantages of reliable and deterministic concept extraction and correlation detection (see Section 5.1.3).

1. **BBQ**: This English-language dataset was developed by Parrish et al. [PCN+22] to expose hidden biases of LLMs against minorities and marginalized groups in a question-answering context. Typically, an ambiguous context with limited additional information is provided, prompting models to rely on hidden cues that often reflect stereotypes. The LLM is required to choose from three answer choices (Person A, Undetermined, Person B), although no objectively correct answer exists. Because this dataset is well suited for exposing biases in LLMs, it was also used by Matton et al. [MNG24] in their own evaluation, making it particularly useful for comparing the behavior of FELICS to that of the original metric.

2. **MedQA**: Introduced by Jin et al. [JPO+21], this dataset poses medical questions to the LLM, drawn from real licensing exams administered in Chinese and English. Like Matton et al. [MNG24], this evaluation only makes use of the subset of English-language questions in which a hypothetical patient is described and the model must select the most likely diagnosis from a set of predetermined answer choices, based on the information provided. Unlike BBQ, MedQA therefore has a clear correct answer; however, this correctness is not used for evaluation, since it is more relevant whether the model's internal reasoning and explanation are aligned than whether its final choice happens to be factually correct.

3. **New German Credit**: The *German Credit* dataset and its updated version, *South German Credit* [Grö19], are widely used within the machine learning community. These datasets consist of tabular information describing credit applicants at a German bank during the 1970s, together with the final decision made by bank employees regarding whether the credit application was approved. One notable aspect of this

data is the correlated nature of several columns. However, the outdated nature of both datasets—for instance, the use of the obsolete *Deutsche Mark* currency, or telephone ownership as a recorded property—may confuse modern LLMs. An updated version inspired by the original dataset was therefore constructed specifically for this evaluation, using an algorithm that reduced the number of columns, made the recorded values more descriptive and up to date, and adjusted monetary amounts such as income to reflect current economic conditions in Germany, while still exhibiting a high degree of correlation among its columns. The resulting dataset is available in the FELICS source code, within the `/data` subfolder.

The general evaluation pipeline described in Section 5 is applied uniformly across all three datasets: input values are modified to observe whether the model's final decision changes. As with the original metric by Matton et al. [MNG24], the prompts used within FELICS still need to be adjusted individually for each dataset; these prompts can likewise be found in the source code, within the `/data` subfolder.

6.2.3 Selection of Models

Causal Concept Faithfulness, as proposed by Matton et al. [MNG24], is designed to be compatible with an arbitrary LLM, a strength that FELICS preserves. Selecting a concrete list of models for evaluation is therefore inherently difficult. This decision was guided by the constraint that a given model must either be hostable on local hardware or accessible via a free API with reasonable usage limits, owing to cost constraints. The following two medium-sized, state-of-the-art LLMs were selected:

1. `gpt-oss:120b` [Ope26]: This LLM, introduced by *OpenAI* as an open-weights model in 2025, employs a mixture-of-experts architecture and comprises 117 billion parameters. According to its developer, its performance is comparable to that of the proprietary `GPT-4o-mini`. It serves as an example of a current American reasoning model and is executed on local infrastructure.

2. `Mistral Medium 3.5` [Mis]: This model has a similar number of parameters (128 billion) but does not employ a *mixture-of-experts* architecture. It was released by the French company *Mistral* in 2026 and is likewise available as an open-weights model with strong reasoning capabilities. The API provided by Mistral offers a generous usage policy, which helps conserve local computing resources.

`gpt-oss:120b` serves not only as a model under evaluation but also as the auxiliary LLM responsible for concept extraction from questions, correlation identification, counterfactual generation, and concept extraction from the generated explanations for all runs of the experiment. This choice is motivated not only by its reliability but also by its similarity to `GPT-4o-mini`, which Matton et al. [MNG24] used as their own auxiliary LLM.

6.2.4 Samples per Question

Matton et al. [MNG24] collected 50 samples per original and counterfactual question in their own work. While this number of samples improves the quality of the resulting estimates, it is infeasible within the time and resource constraints of this thesis.

Therefore, for both models, $n = 10$ responses are gathered for *BBQ* and *MedQA*. Although this constitutes only a fifth of the sample size used by Matton et al. [MNG24], it is sufficient to obtain meaningful results. For *New German Credit*, the number of samples can be further reduced to $n = 5$, since each question is conceptually identical and differs only in its underlying data, meaning that fewer data points are required.

6.2.5 Removal vs. Value-Replacement Counterfactuals

For MedQA, Matton et al. [MNG24] use counterfactuals in which the value of the concept under intervention is *removed*. They argue that, within a medical context, verifying the plausibility of alternative values generated by the auxiliary LLM is difficult for non-experts, and that determining a meaningful alternative value is not always possible for certain factors (e.g., blood pressure). Conversely, for the experiments on the BBQ dataset, Matton et al. [MNG24] use *value replacements*, in which properties are typically swapped between the two persons described in the question.

Although it would be interesting to evaluate alternative intervention approaches, it has been decided, for the sake of comparability and given the limited available resources, that the evaluation is conducted using exactly the same counterfactual-generation approach as Matton et al. for each respective dataset. The newly introduced New German Credit dataset is assessed using *concept removal*, since the absence of potentially conflicting information (as discussed in Section 5.5.3) may help to better reveal the redundant interaction effects of certain concepts.

6.3 Evaluation Metrics

Whereas Section 6.1 states *what* each hypothesis claims, this section defines *how* each claim is tested against the outputs of the reference implementation, using the experimental settings established above. Outputs from *Causal Concept Faithfulness* by Matton et al. [MNG24] and from *FELICS* are analyzed using the procedures listed below within the statistical software *R*. The corresponding scripts can be found within the `/evaluation` subfolder of the FELICS repository, as introduced in Section 6.2.1.

6.3.1 Evaluation of RQ1: Correlation-Strength Impact

The relationship between concept correlation and the resulting CE difference is assessed on the natural language datasets, comparing an atomic ($n_{\max} = 1$) against a joint ($n_{\max} = 4$) configuration of FELICS.

For each dataset $X \in \{\text{BBQ}, \text{MedQA}\}$ and evaluated model M: for every question x with concept set C, define the correlation strength under configuration $n_{\max}$ as

$$\kappa(x, n_{\max}) := \frac{1}{|C|} \sum_{s \in \mathcal{O}(x, C, n_{\max})} \sum_{\emptyset \neq T \subseteq s} |T|,$$

i.e., the total number of concept-occurrences across all treatments evaluated for x, normalized by the number of concepts $|C|$. Note that $\kappa(x, 1) = 1$ for every question, since every correlation group is a singleton under the atomic configuration. Let

$\Delta\kappa(x) := |\kappa(x,4) - \kappa(x,1)|$ denote the difference in correlation strength between the joint and atomic configurations, and let $\Delta CE(x) := |\overline{CE}_{\text{adj}}(x,4) - \overline{CE}_{\text{adj}}(x,1)|$ denote the absolute difference between the mean adjusted Shapley CE across all concepts of x, computed under both configurations. Assemble a table with one row per question containing $\Delta\kappa(x)$ and $\Delta CE(x)$, and compute the Pearson correlation between both columns using a one-sided test (alternative: greater). The hypothesis is considered supported for a given dataset and model if $r \geq 0.10$ and $p \leq 0.10$. Repeat the procedure independently for each dataset and model, and, in a second pass, substitute the question-level faithfulness score $F(x)$ for $\overline{CE}_{\text{adj}}(x, n_{\max})$ to test the analogous effect on faithfulness. Fixing the joint configuration to $n_{\max} = 4$ for BBQ and MedQA is a reasonable choice: a smaller value would restrict the oracle's ability to form larger correlation groups and thus bias the analysis toward the null hypothesis, whereas $n_{\max}$ must nonetheless be kept identical across all questions of a dataset to keep $\Delta\kappa(x)$ comparable. This hypothesis cannot be tested on New German Credit, however, since its oracle produces a single, dataset-wide partition of concepts (Section 5.1.3) rather than a per-question grouping. Consequently, $\kappa(x, n_{\max})$ would be identical for every question in the dataset, leaving no variance in $\Delta\kappa(x)$ against which to correlate $\Delta CE(x)$.

6.3.2 Evaluation of RQ2: Baseline Equivalence

The convergence of FELICS ($n_{\max} = 1$) to the original *Causal Concept Faithfulness* baseline is evaluated using an equivalence-testing rather than a difference-testing procedure, since the goal is to demonstrate the absence of a meaningful difference rather than the presence of one. Because the original metric does not support tabular data, this hypothesis is restricted to the BBQ and MedQA datasets.

For each dataset $X \in \{\text{BBQ}, \text{MedQA}\}$ and evaluated model M: restrict both the Matton et al. and the FELICS ($n_{\max} = 1$) outputs to the set of questions present in both. For each shared question x, compute the mean CE across all its concepts—using $CE_{\text{Matton}}(x, C_m)$ for the baseline and $CE_{\text{adj}}(x, C_m, \{C_m\})$ for FELICS—as well as the question-level faithfulness score $F(x)$ obtained from each pipeline. Conduct a paired Two One-Sided Tests (TOST) procedure for both the CE sample and the faithfulness sample, using equivalence bounds of $\pm 25\%$ and $\pm 10\%$ of the standard deviation of the paired differences, respectively. The hypothesis is considered supported for a given dataset and model if both one-sided p-values of the TOST procedure fall below 0.20, for both the CE comparison and the faithfulness comparison.

6.3.3 Evaluation of RQ3: Rank Preservation

Rank preservation is evaluated on the BBQ, MedQA, and New German Credit datasets under both an atomic and a joint oracle configuration.

For each dataset $X \in \{\text{BBQ}, \text{MedQA}, \text{New German Credit}\}$, evaluated model M, and oracle configuration—$n_{\max} = 1$, included as a sanity check since CE_{raw} and CE_{adj} coincide trivially for singleton groups, and the joint configuration ($n_{\max} = 4$ for BBQ and MedQA, $n_{\max} = 3$ for New German Credit)—collect, for every concept-level observation, the raw Shapley causal effect

$$CE_{\text{raw}}(x, C_m, s) := \phi_m(v)$$

obtained from Equation 5.7 using the unadjusted value function v, alongside the adjusted Shapley causal effect $CE_{\text{adj}}(x, C_m, s)$ obtained from Equation 5.24. Compute the Spearman rank correlation ρ between both columns using a one-sided test (alternative: greater). The hypothesis is considered supported for a given dataset, model, and configuration if $\rho \geq 0.9$ and $p \leq 0.05$.

6.3.4 Evaluation of RQ4: Faithfulness Enhancement

The effect of modeling interactions on faithfulness is evaluated on all datasets and models, comparing the atomic baseline against a joint configuration.

For each dataset $X \in \{\text{BBQ}, \text{MedQA}, \text{New German Credit}\}$ and evaluated model M: pair each question's atomic ($n_{\max} = 1$) question-level faithfulness score with its joint-configuration counterpart ($n_{\max} = 4$ for the natural-language datasets BBQ and MedQA, $n_{\max} = 3$ for the tabular New German Credit dataset, consistent with the maximum group sizes established for the respective oracle implementations in Section 5.1). Conduct a paired, one-sided Wilcoxon Signed-Rank Test with the alternative hypothesis that the joint-configuration faithfulness scores are stochastically greater than the atomic ones. The hypothesis is considered supported for a given dataset and model if the resulting p-value is ≤ 0.05.

6.3.5 Evaluation of RQ5: Motivating-Example Resolution

The resolution of this specific case is verified through a targeted, qualitative analysis of the corresponding MedQA question, repeated for each evaluated model.

For each evaluated model M: run the identified MedQA question through both the original and the FELICS pipeline, and construct a table with columns *Concept*, CE_{original}, and CE_{adj}, listing every concept present in the question. Isolate the two concepts of interest—*the patient's eating disorder* and *the patient's self-perception of weight*—and assert that $CE_{\text{adj}} > CE_{\text{original}}$ holds for both. Because this hypothesis concerns a single, specific question rather than a distribution of questions, no inferential statistical test is applicable; the analysis is instead reported as a qualitative case study, and the assertion is required to hold for every evaluated model.

Chapter 7
Evaluation Methodology

7.1 Research Questions

Defining research questions and hypothesis like "does FELICS behave in the way we expect it".

7.2 Datasets

Explain datasets and why they are chosen for evaluation: 1. BBQ, 2. MedQA, 3. NewGermanCredit

7.3 Evaluated Models

Explain which models and why I am testing them. Also talk briefly about constraints that led to the decision.

7.4 Reference Implementation

Explain things about the actual implementation, where to find it (Github) and that its essentially a fork from Mattons original. Also explain that we also run Mattons implementation as comparison with the same settings and where to find their implementation (also on GitHub).

7.5 Baselines

What am I comparing against: Once FELICS with correlation enabled ($max_i n_g roup = 4$), FELICS with correlation disabled ($max_i b_g roup = 1$) and Mattons original.

7.6 Evaluation Metrics

How am I verifying my questions and hypothesis? E.g. conducting a t-test on results in R to check for significant changes, to practically prove backwards compatibility.

Chapter 8

Results

The most pressing results in this chapter. Move detailed views and large tables into Appendix. Therefore keep this short, max 3 pages. Just mention observations, but keep explanations or discussions for later.

Chapter 9
Discussion

9.1 Interpretation

Look based on the data if research questions can be answered and what the result is. What might have led to these results?

9.2 Desiderata Revisited

Answer based on the results. From a practical standpoints, were the desiderata fulfilled? Don't make mathematical proves in this chapter, the mathematical/logical aspects belong in "Modeling".

9.3 Limitations in Experimental Setup

What was not great about the evaluation (e.g. small sample size, only few combinations tested).

9.4 Limitations of Metric

Which technical limitations does the metric have? E.g. getting rid of the CE-Normalization (Payout Coefficient) for a Bayesian aproach or even choosing to break compatibility for better results. Be fair, but don't discredit my own work too much since this would pretty much destroy my thesis.

9.5 Future Work

Discuss what could be addressed in future work and what more needs to be researched?

Chapter 10
Conclusion

Finish the entire thesis off in around 2, max 3 pages.

Appendix A
Examples and Demonstrations

A.1 Effects of Normalization on Pearson Correlation

Example A.1

Demonstration: Payout normalization can alter the Pearson correlation between *CE* and *EE* vectors.

Setup: Consider a minimal dataset with three questions x_1, x_2, x_3 and five concepts $C_1, \ldots, C_5$. The oracle groups concepts as follows:

- x_1: $s_1 = \{C_1, C_2\}$ (redundant pair, $v(\{C_1, C_2\}) = 1.5 < v(\{C_1\}) + v(\{C_2\}) = 2$)
- x_2: $s_2 = \{C_3, C_4\}$ (less redundant pair, $v(\{C_3, C_4\}) = 1.8 < v(\{C_3\}) + v(\{C_4\}) = 2$)
- x_3: $s_3 = \{C_5\}$ (singleton)

The value functions for all singleton treatments satisfy $v(\{C_m\}) = 1$ for $m \in \{1, 2, 3, 4\}$ and $v(\{C_5\}) = 2$. The reference set for size-1 treatments is $\mathcal{R}_1 = \{\{C_1\}, \{C_2\}, \{C_3\}, \{C_4\}, \{C_5\}\}$, yielding

$$\gamma(x, \{C_m\}) = \frac{v(\{C_m\})}{v(\{C_m\})} = 1 \quad \Rightarrow \quad \bar{\gamma}(\cdot, \cdot, \{C_m\}) = 1.$$

For size-2 treatments, $\mathcal{R}_2 = \{\{C_1, C_2\}, \{C_3, C_4\}\}$ gives

$$\bar{\gamma}(\cdot, \cdot, \{C_1, C_2\}) = \bar{\gamma}(\cdot, \cdot, \{C_3, C_4\}) = \frac{1}{2}\left(\frac{1.5}{2} + \frac{1.8}{2}\right) = 0.825.$$

Original Metric (Matton et al.): All concepts are treated as singletons, so

$$\boldsymbol{CE}_{\text{original}} = [\, v(\{C_1\}),\, v(\{C_2\}),\, v(\{C_3\}),\, v(\{C_4\}),\, v(\{C_5\}) \,] = [1,\, 1,\, 1,\, 1,\, 2]\,.$$

FELICS with Normalization: For $s_1 = \{C_1, C_2\}$ and $s_2 = \{C_3, C_4\}$, the normalized value functions are

$$\begin{aligned} v_{\text{adj}}(\{C_1\}) &= v(\{C_1\})/1 = 1, \\ v_{\text{adj}}(\{C_1, C_2\}) &= v(\{C_1, C_2\})/0.825 \approx 1.818. \end{aligned}$$

By the Shapley efficiency axiom, $\sum_{C_m \in s} CE_{\text{adj}}(x, C_m, s) = v_{\text{adj}}(s)$. Thus, for s_1:

$$\begin{aligned} CE_{\text{adj}}(x_1, C_1, s_1) &= 0.5 \cdot (v_{\text{adj}}(\{C_1\}) - 0) + 0.5 \cdot (v_{\text{adj}}(\{C_1, C_2\}) - v_{\text{adj}}(\{C_2\})) \\ &= 0.5 \cdot 1 + 0.5 \cdot (1.818 - 1) \approx 0.909, \\ CE_{\text{adj}}(x_1, C_2, s_1) &\approx 0.909, \end{aligned}$$

and analogously $CE_{\text{adj}}(x_2, C_3, s_2) = CE_{\text{adj}}(x_2, C_4, s_2) \approx 1.091$. For the singleton s_3, $CE_{\text{adj}}(x_3, C_5, s_3) = v_{\text{adj}}(\{C_5\}) = 2$. Hence,

$$\boldsymbol{CE}_{\text{adj}} = [0.909,\ 0.909,\ 1.091,\ 1.091,\ 2]\,.$$

Explanation-Implied Effects: Assume the LLM's explanations yield

$$\boldsymbol{EE} = [1.0,\ 0.5,\ 1.0,\ 1.0,\ 2.0]\,.$$

Pearson Correlation Computation: For both vectors, $\mu_{\boldsymbol{EE}} = 1.1$. The original metric gives

$$\begin{aligned}
\mu_{\boldsymbol{CE}_{\text{original}}} &= \frac{1+1+1+1+2}{5} = 1.2,\\
\text{cov}(\boldsymbol{CE}_{\text{original}}, \boldsymbol{EE}) &= \frac{1}{5}\sum_{i=1}^{5}(CE_{\text{original},i} - 1.2)(EE_i - 1.1) = 0.9,\\
\text{var}(\boldsymbol{CE}_{\text{original}}) &= \frac{1}{5}\sum_{i=1}^{5}(CE_{\text{original},i} - 1.2)^2 = 0.8,\\
\text{var}(\boldsymbol{EE}) &= \frac{1}{5}\sum_{i=1}^{5}(EE_i - 1.1)^2 = 1.19,\\
\text{PCC}_{\text{original}} &= \frac{0.9}{\sqrt{0.8 \cdot 1.19}} \approx 0.923.
\end{aligned}$$

For FELICS, $\mu_{\boldsymbol{CE}_{\text{adj}}} = 1.2$ (dataset-level average preserved), but

$$\begin{aligned}
\text{cov}(\boldsymbol{CE}_{\text{adj}}, \boldsymbol{EE}) &= \frac{1}{5}\sum_{i=1}^{5}(CE_{\text{adj},i} - 1.2)(EE_i - 1.1) \approx 0.9455,\\
\text{var}(\boldsymbol{CE}_{\text{adj}}) &\approx 0.8332,\\
\text{PCC}_{\text{adj}} &\approx \frac{0.9455}{\sqrt{0.8332 \cdot 1.19}} \approx 0.946.
\end{aligned}$$

Conclusion: Although $\mu_{\boldsymbol{CE}_{\text{adj}}} = \mu_{\boldsymbol{CE}_{\text{original}}}$ (dataset-level backward compatibility), the covariance and variance terms differ due to the non-uniform scaling of $\boldsymbol{CE}$ across group boundaries. Thus, $\text{PCC}_{\text{adj}} \neq \text{PCC}_{\text{original}}$, proving that the faithfulness score is not guaranteed to be preserved.

A.2 CE Dataset-Average Preservation

Mathematical Demonstration A.1

Claim: The dataset-level average of all Causal Effect values computed by FELICS is *approximately* equal to that of Matton et al.'s original metric:

$$\frac{1}{N_C}\sum_{x\in X}\sum_{C_m\in C(x)} CE_{\text{adj}}(x, C_m, s) \approx \frac{1}{N_C}\sum_{x\in X}\sum_{C_m\in C(x)} v(\{C_m\}).$$

Here, N_C is the total number of concepts, $C(x)$ are the concepts in question x, and s is the correlation group containing C_m.

Intuition: The payout-normalization step is designed to compensate for the *soft ceiling effect* (Section 5.3.1), which systematically reduces joint effects $v(s)$ relative to the sum of singleton effects $\sum_{C_m \in s} v(\{C_m\})$. If this compensation is applied consistently across the dataset, the *total causal influence* attributed to all concepts should remain approximately unchanged, preserving the overall scale of CE values.

Simplifying Assumptions for Demonstration: To illustrate why approximate equality is plausible, two idealized conditions will be relied on that do not hold universally but capture the *typical* behavior of the reference-set construction:

1. **Representative Reference Sets:** The reference set $\mathcal{R}(x, s, s)$ (Equations 5.18–5.20) is treated as a *representative sample* of all groups of size $|s|$ in the dataset. In practice, $\mathcal{R}(x, s, s)$ contains at most 5 elements, drawn prioritized by proximity (same group $\to$ same question $\to$ same dataset). For large datasets, this sample is approximately representative of the population of size-$|s|$ groups.

2. **Symmetry of Payout Coefficients:** The payout coefficients $\gamma(x', s')$ for groups of the same size are are assumed to be *exchangeable*, i.e., their distribution is invariant to permutation. This allows us to treat $\bar{\gamma}(x, s, s)$ as an estimate of a global average $\bar{\gamma}_k$ for size-k groups.

The assumptions above are sufficient to justify the *approximate* preservation expected to be observed in practice.

Demonstration Under Assumptions: For any group s, the Shapley *efficiency axiom* gives

$$\sum_{C_m \in s} CE_{\text{adj}}(x, C_m, s) = v_{\text{adj}}(s) = \frac{v(s)}{\bar{\gamma}(x, s, s)}. \tag{A.1}$$

Under Assumption (1), $\bar{\gamma}(x, s, s) \approx \bar{\gamma}_k$, where $\bar{\gamma}_k$ is the average payout coefficient for all size-k groups:

$$\bar{\gamma}_k := \frac{1}{|\mathcal{G}_k|} \sum_{s' \in \mathcal{G}_k} \frac{v(s')}{\sum_{C_m \in s'} v(\{C_m\})}.$$

Here, $\mathcal{G}_k$ denotes the set of all size-k groups (distinct from the graph G in Section 5.1). Substituting into Equation A.1:

$$\begin{aligned}
\sum_{s \in \mathcal{G}_k} \sum_{C_m \in s} CE_{\text{adj}}(x, C_m, s) &= \sum_{s \in \mathcal{G}_k} \frac{v(s)}{\bar{\gamma}(x, s, s)} \approx \sum_{s \in \mathcal{G}_k} \frac{v(s)}{\bar{\gamma}_k} \\
&= \frac{1}{\bar{\gamma}_k} \sum_{s \in \mathcal{G}_k} v(s) = \frac{1}{\bar{\gamma}_k} \cdot \bar{\gamma}_k \sum_{s \in \mathcal{G}_k} \sum_{C_m \in s} v(\{C_m\}) \\
&= \sum_{s \in \mathcal{G}_k} \sum_{C_m \in s} v(\{C_m\}).
\end{aligned}$$

The second equality follows from the definition of $\bar{\gamma}_k$: multiplying $\bar{\gamma}_k$ by the sum of all $v(s)$ for $s \in \mathcal{G}_k$ yields the sum of all $\sum_{C_m \in s} v(\{C_m\})$. Summing over all group sizes k:

$$\begin{aligned}\sum_{x\in X}\sum_{s\in S(x)}\sum_{C_m\in s} CE_{\text{adj}}(x, C_m, s) &\approx \sum_{x\in X}\sum_{s\in S(x)}\sum_{C_m\in s} v(\{C_m\}) \\ &= \sum_{x\in X}\sum_{C_m\in C(x)} v(\{C_m\}).\end{aligned}$$

Dividing by N_C preserves the approximation, showing that the dataset-level averages are approximately equal. The error introduced by the simplifications diminishes as the dataset grows and the reference sets become more representative.

Appendix B
Prompts

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